



Antimycobacterial drugs

21 January 2026, 11:00-12:00
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Objectives

After this class, students should be able to:

- explain the mechanisms of action of antimycobacterial drugs.
- describe the pharmacokinetics & pharmacodynamics, and spectrum of activity of antimycobacterial drugs.
- identify common adverse drug reactions and drug interactions of antimycobacterial drugs.



Mycobacteria

Key characteristics:

- aerobic, rod-shaped, non-motile bacilli
- thick, hydrophobic, mycolic acid-rich cell wall

Common mycobacterial infections

- Tuberculosis (TB)
- Leprosy
- Nontuberculous mycobacteria (NTM)



Antimycobacterial drugs

Drugs used in Tuberculosis

- Isoniazid (H,INH)
- Rifampin (R)
- Pyrazinamide (Z)
- Ethambutol (E)
- Fluoroquinolones
- Aminoglycosides
- Ethionamide
- Cycloserine
- Para-aminosalicylic acid (PAS)
- Clofazimine
- Bedaquiline
- Delamanid
- Protomanid

Drugs used in Leprosy

- Dapsone
- Clofazimine
- Rifampin

Drugs used in non-tuberculous mycobacteria

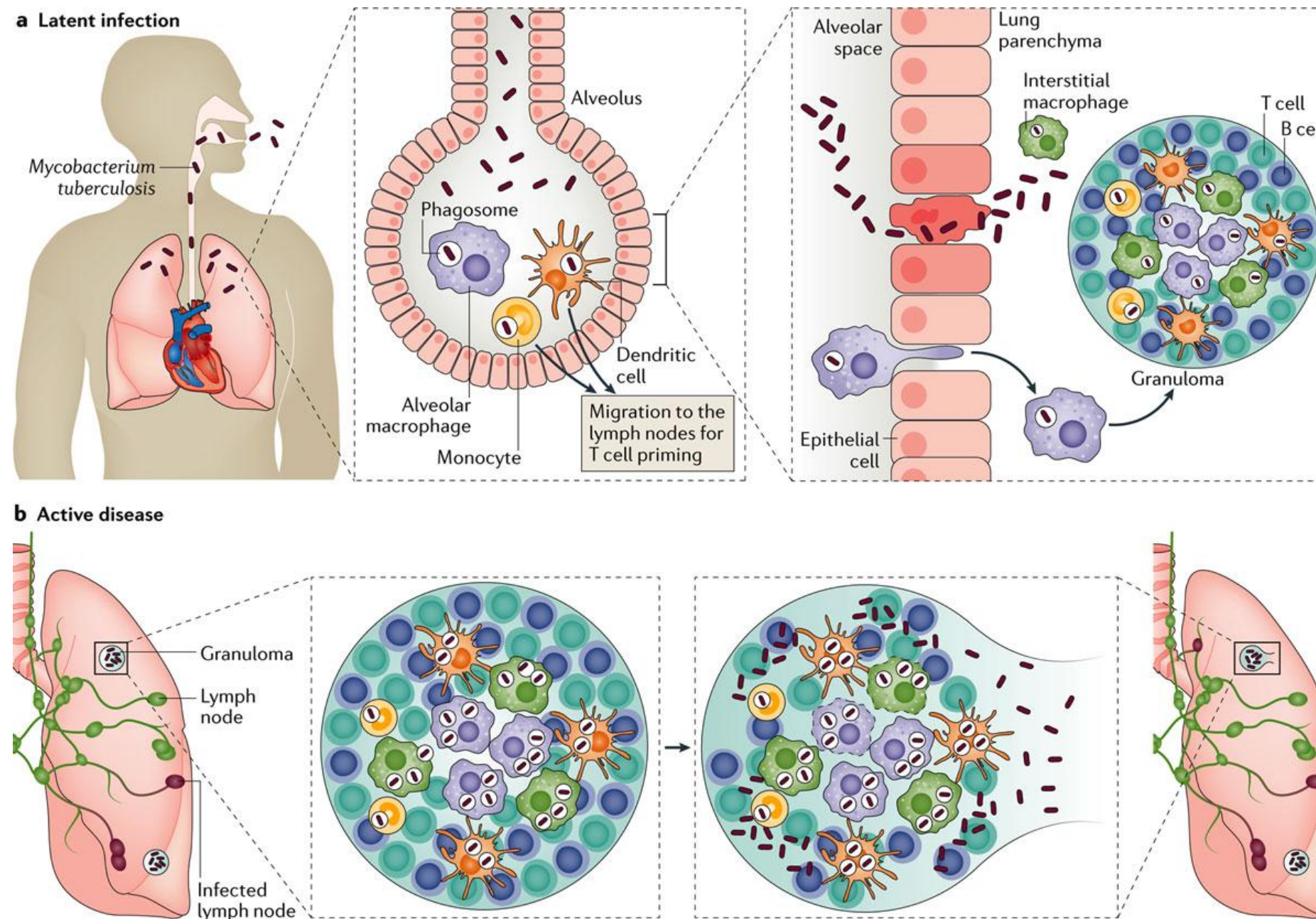
- Macrolides
- Fluoroquinolones
- Aminoglycosides



Tuberculosis (TB)

- *Mycobacterium tuberculosis* (*M. tuberculosis*)
 - Weakly gram-positive, aerobic bacillus, rod-shaped
 - Non-motile, non-sporulating
 - Transmission: airborne droplets
 - Slow-growing (doubling time of 18–24 hrs.)
 - Cell wall content: rich in high-MW lipids
 - Intrinsically resistant to many drugs
- Need co-administrations of several drugs with different mechanisms.
- Slow drug response requiring prolonged therapy.

Mycobacterium tuberculosis infection



Nature Reviews | Disease Primers

Pai, M. et al. (2016) Tuberculosis
Nat. Rev. Dis. Primers doi:10.1038/nrdp.2016.76

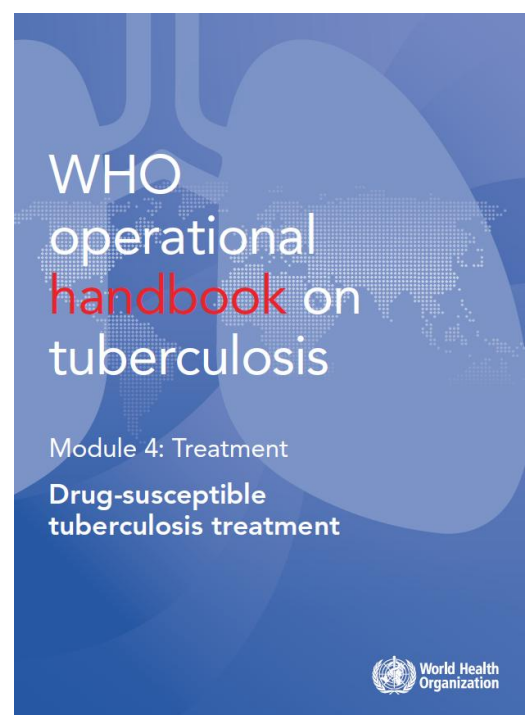


Why are mycobacterial infections difficult to treat?

- **Waxy cell wall:** composed of $> 60\%$ lipids, mainly mycolic acids forming a protective barrier that limits drugs penetration into the cell.
- **Abundant efflux pumps:** e.g. ATP-binding cassette (ABC) transporters, actively expel antimicrobial agents, contributing to intrinsic drug resistance.
- **Location in host:** reside both intracellularly & extracellularly, requiring drugs with good tissue & intracellular penetration.



TB treatment guidelines



Check for updates

AMERICAN THORACIC SOCIETY DOCUMENTS

Updates on the Treatment of Drug-Susceptible and Drug-Resistant Tuberculosis

An Official ATS/CDC/ERS/IDSA Clinical Practice Guideline

© Jussi J. Saukkonen*, Raquel Duarte*, Sonal S. Munsiff*, Carla A. Winston*, Manoj J. Mammen, Ibrahim Abubakar, Carlos Acuña-Villaorduña, Pennan M. Barry, Mayara L. Bastos, Wendy Carr, Hassan Chami, Lisa L. Chen, Terence Chorbha, Charles L. Daley, Anthony J. Garcia-Prats, Kelly Holland, Ioannis Konstantinidis, Marc Lipman, Giovanni Battista Migliori, Farah M. Parvez, Adrienne E. Shapiro, Giovanni Sotgiu, Jeffrey R. Starke, Angela M. Starks, Sanket Thakore, Shu-Hua Wang, Jonathan M. Wortham, and Payam Nahid; on behalf of the American Thoracic Society, U.S. Centers for Disease Control and Prevention, European Respiratory Society, and Infectious Diseases Society of America

THIS OFFICIAL CLINICAL PRACTICE GUIDELINE WAS APPROVED BY THE AMERICAN THORACIC SOCIETY (ATS) AND THE INFECTIOUS DISEASES SOCIETY OF AMERICA (IDSA) SEPTEMBER 2024, WAS CLEARED BY THE U.S. CENTERS FOR DISEASE CONTROL AND PREVENTION (CDC) SEPTEMBER 2024, AND WAS APPROVED BY THE EUROPEAN RESPIRATORY SOCIETY (ERS) OCTOBER 2024

Abstract

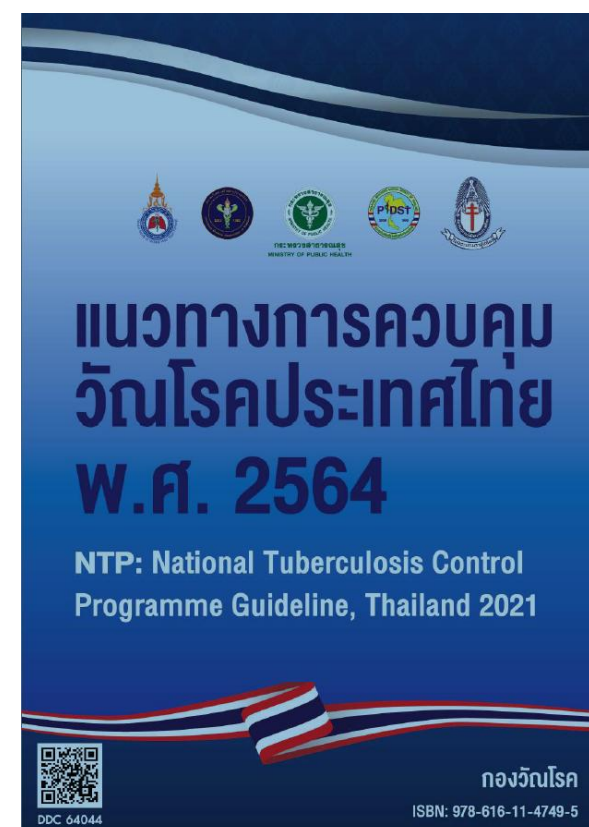
Background: On the basis of recent clinical trial data for the treatment of drug-susceptible and drug-resistant tuberculosis (TB), the American Thoracic Society, U.S. Centers for Disease Control and Prevention, European Respiratory Society, and Infectious Diseases Society of America have updated clinical practice guidelines for TB treatment in children and adults in settings in which mycobacterial cultures, molecular and phenotypic drug susceptibility tests, and radiographic studies, among other diagnostic tools, are available on a routine basis.

Methods: A Joint Panel representing multiple interdisciplinary perspectives convened with American Thoracic Society methodologists to review evidence and make recommendations using the GRADE (Grading of Recommendations Assessment, Development and Evaluation) and GRADE-ADOLOPMENT (adoption, adaptation, and, as needed, *de novo* development of recommendations) methodology.

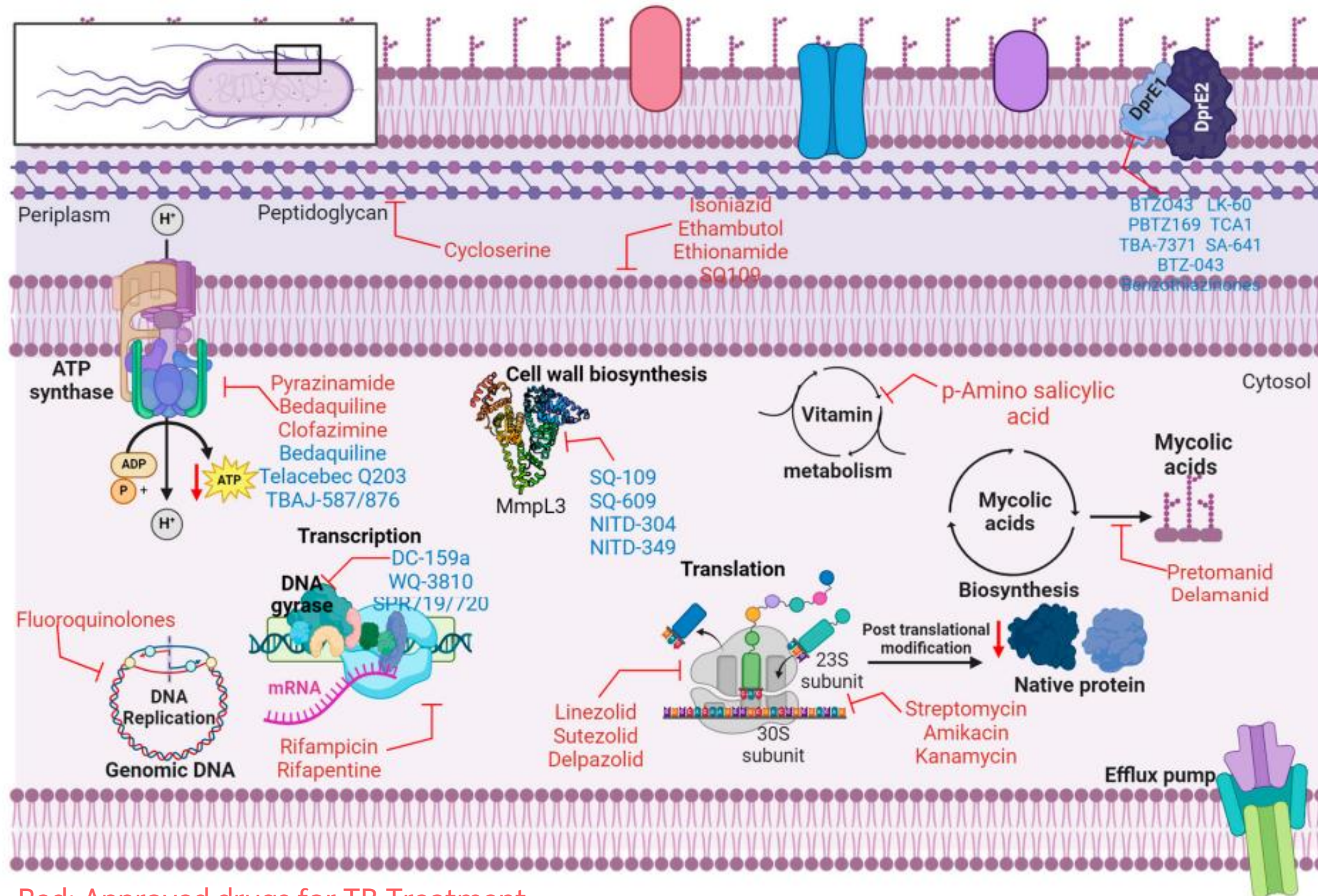
Results: New drug-susceptible TB recommendations include the use of a novel 4-month regimen for people with pulmonary TB and a shortened 4-month regimen for children with nonsevere TB. Drug-resistant TB recommendation updates include the use of novel regimens containing bedaquiline, pretomanid, and linezolid with or without moxifloxacin.

Conclusions: All-oral, shorter treatment regimens for TB are now recommended for use in eligible individuals.

Keywords: tuberculosis; drug-resistant; drug-susceptible; children; adults



Mechanisms of action of anti-tuberculosis drugs



Red: Approved drugs for TB Treatment
Blue: Investigational drugs

Microorganisms 2025, 13, 722.



Drug-susceptible TB: Active

- 6-month regimen

“2HRZE / 4HR” - 2 months – Intensive phase
- 4 months – Continuous phase

Isoniazid (H), Rifampin (R), Pyrazinamide (Z), Ethambutol (E)

- 4-month regimen

“2HPMZ / 2HPM” - 2 months – Intensive phase
- 2 months – Continuous phase

Isoniazid (H), Rifapentine (P), Pyrazinamide (Z), Moxifloxacin (M)

Directly Observed Therapy (DOT)

Intensive phase - kills 99% of the bacterial population.
- significantly reduces bacterial loads & symptoms.

Continuation phase - sterilizes remaining bacteria, preventing relapse.

Overall cure rate: >85% for drug-susceptible TB with proper adherence.



Drug-susceptible TB: Latent

- **Latent TB infection**
 - asymptomatic, non-infectious
 - 5-10% of people with latent TB can develop active TB disease.
 - ↑ Risk: immunocompromised conditions.
- **Treatment regimens** (kill dormant bacteria)
 - **3HP** (Isoniazid + Rifapentine): once weekly for 3 months (12 doses)
 - **4R** (Rifampin): once daily for 4 months
 - **3HR** (Isoniazid + Rifampin): once daily for 3 months
 - **6H** (Isoniazid): daily or twice weekly for 6 months
 - **9H** (Isoniazid): daily for 9 months



Isoniazid

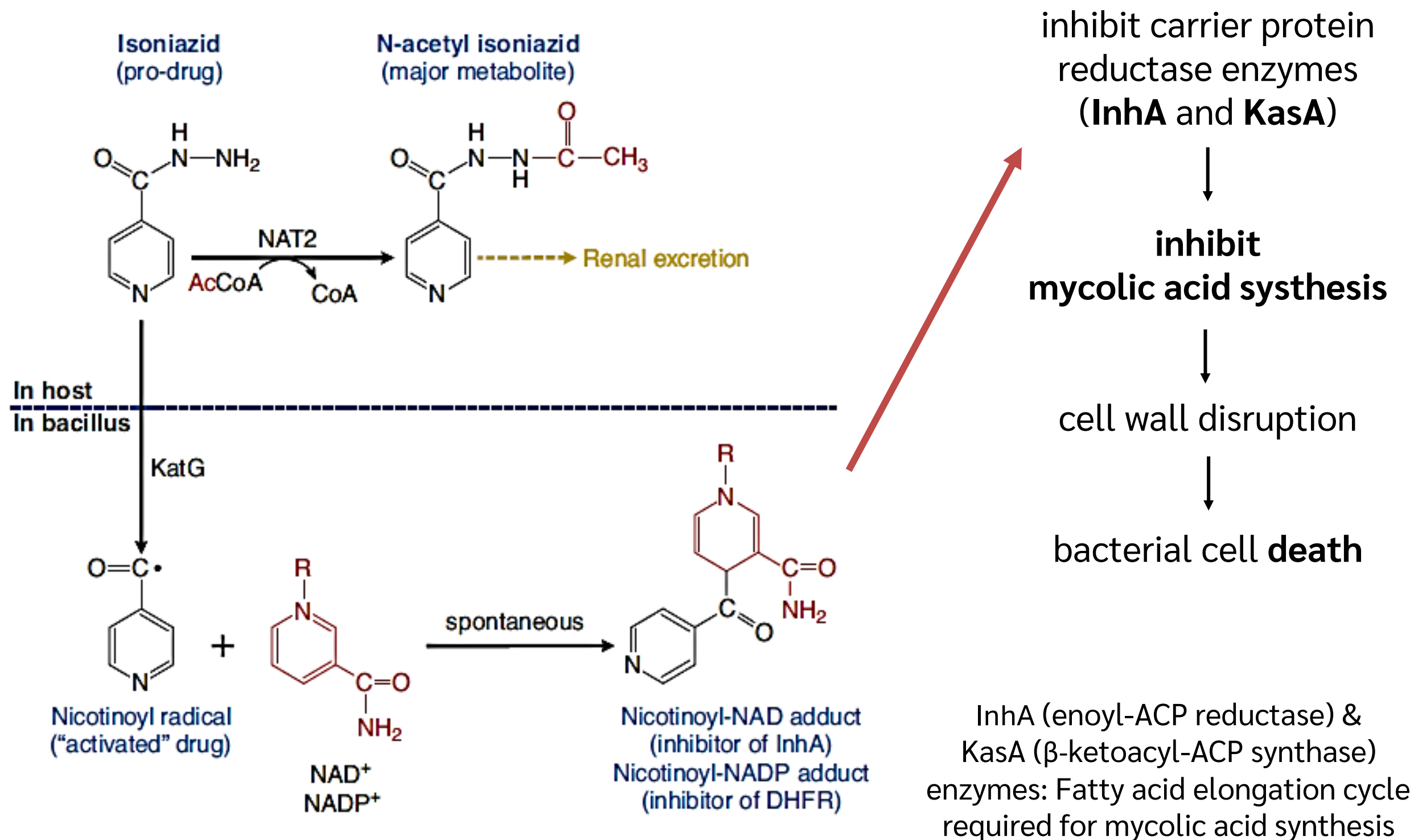
Mechanism of action:

- inhibits synthesis of mycolic acids.
 - bactericidal against actively dividing bacteria.
 - bacteriostatic against dormant bacteria.
-
- Highly effective in killing *M. tuberculosis* during the initial intensive phase of therapy.

Indications:

- Active & Latent tuberculosis infection

Isoniazid



KatG: catalase-peroxidase (KatG)



Isoniazid

Resistance:

- Resistance can emerge rapidly if the drug is used alone.
- Most common & most severe:
 - *katG* gene mutation (reducing prodrug activation)
 - usually co-occurrence with ethambutol resistant
- Less common:
 - *inhA* gene mutation (altering the target enzyme)
 - *KasA* gene mutation (altering the target enzyme)



Isoniazid

Pharmacokinetics:

- **Absorption:** bioavailability ~ 90-100% (food reduces bioavailability)
- **Distribution:** High Vd, distributes widely to all body tissues & fluids (including CSF)
- **Metabolism:** hepatic via N-acetylation

Genetic polymorphisms in N-acetyltransferase 2 (NAT2):

- *Fast acetylators: half-life: 1- 1.5 hrs* (Inuit, Japanese)
- *Slow acetylators: half-life: 2-5 hrs* (Scandinavians, Jews, North African whites, Thai (30%))
- **Elimination:** via urine, mostly as inactive metabolites, 75–95% of the dose is excreted within 24 hrs.

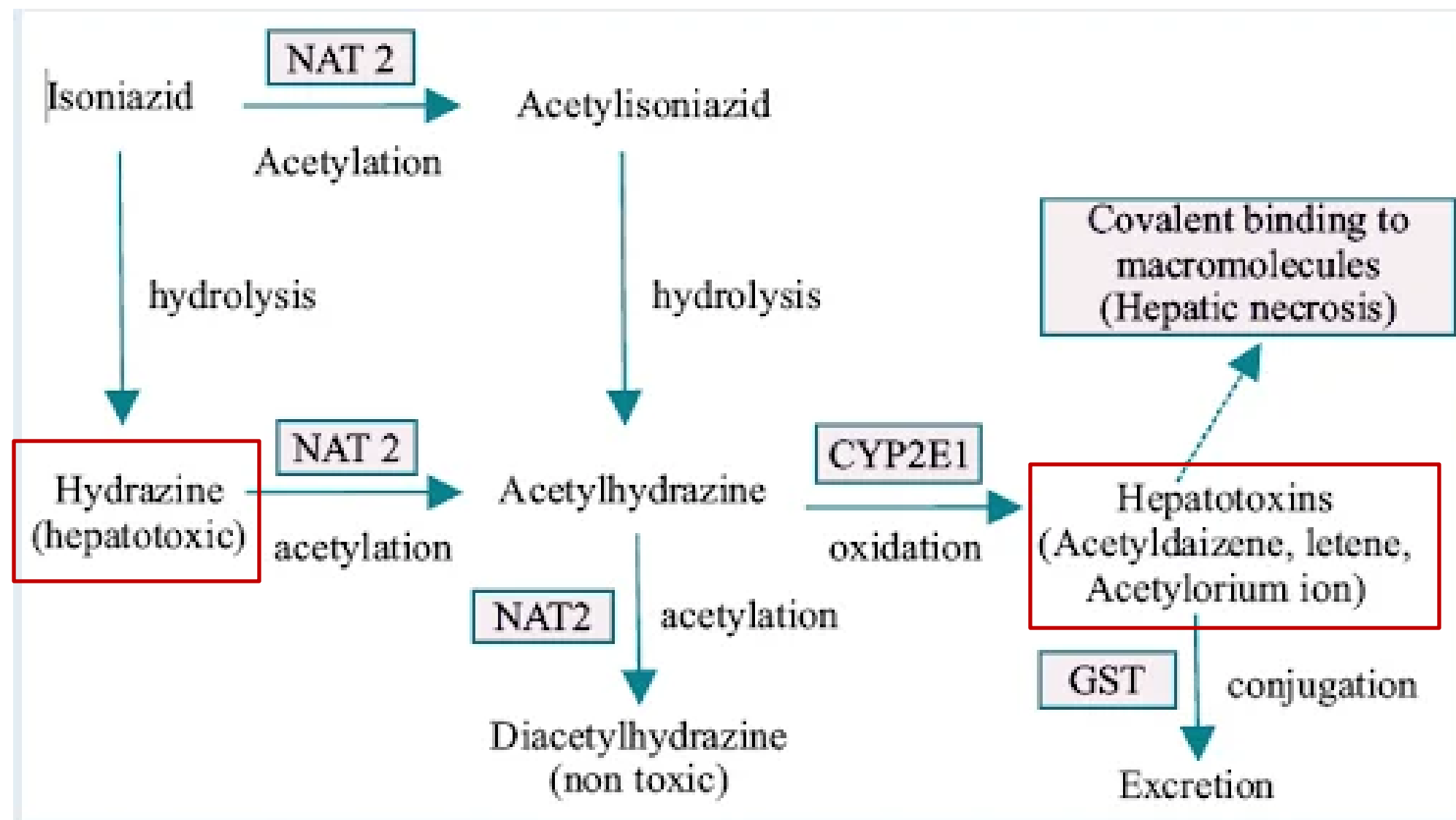


Isoniazid

Adverse drug reactions:

- Hepatotoxicity
 - Asymptomatic, $\uparrow$ AST, $\uparrow$ ALT (common).
 - Hepatitis (rare but serious, especially in older adults & heavy alcohol users).
 - Risk Factors:
 - Age >35 years.
 - Slow acetylators
 - Co-administration with rifampin or alcohol use
 - Underlying liver disease

INH-induced hepatotoxicity



- Fast acetylators → ↓INH serum level
- Slow acetylators → ↑INH serum level → ↑hepatotoxicity risk
- Less amount of NAT2 → ↑hepatotoxicity risk



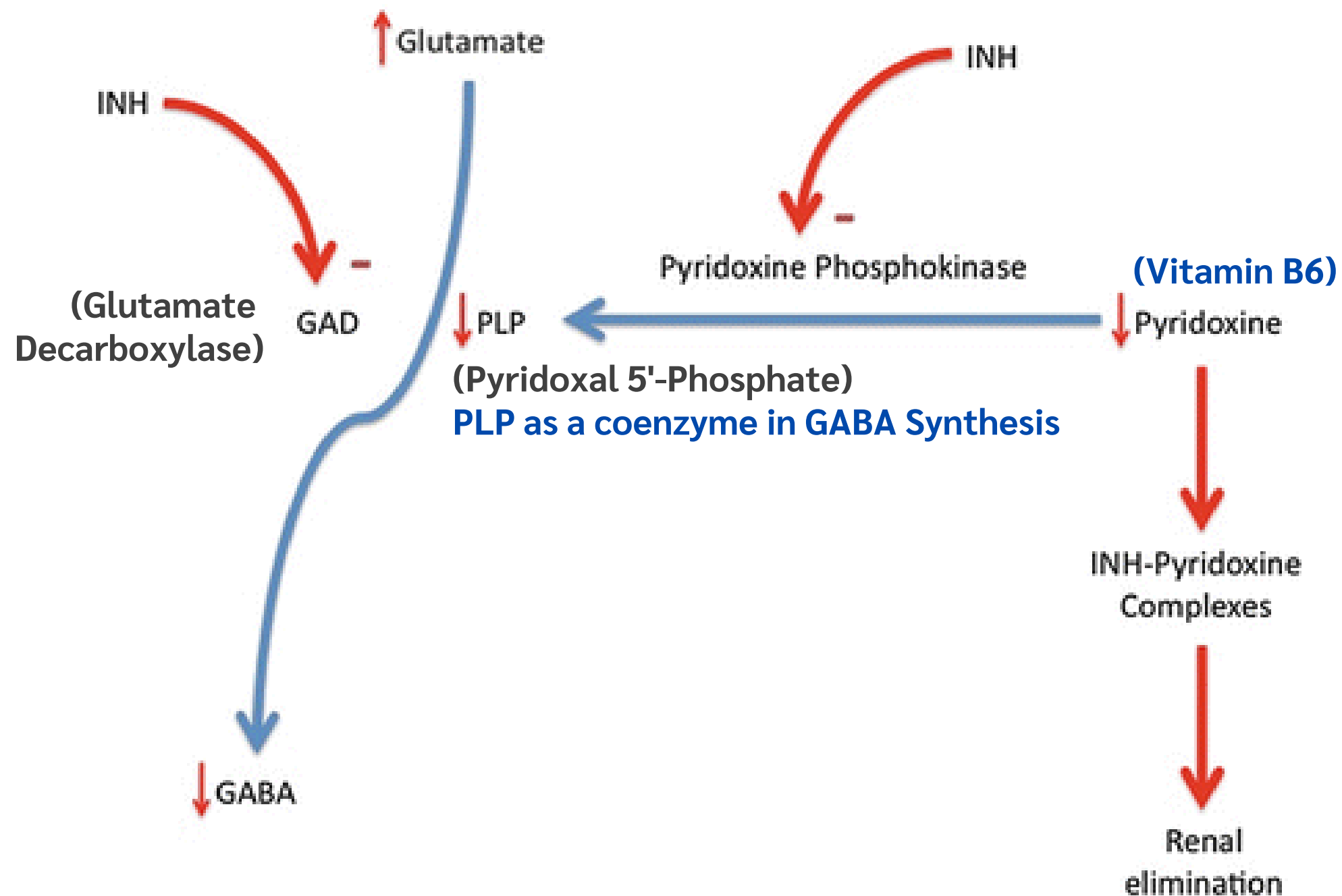
Isoniazid

Adverse drug reactions:

- Peripheral neuropathy (most common)
 - caused by INH-induced depletion of pyridoxine (vitamin B6).
 - symptoms: tingling, numbness, or burning sensations in hands & feet.
 - risk factors: malnutrition, diabetes, alcoholism, pregnancy, or slow acetylators.
 - prevention: pyridoxine supplementation (10–50 mg/day)

Routine co-administration of pyridoxine (10–50 mg/day) with INH is standard practice to prevent peripheral neuropathy.

INH-induced peripheral neuropathy



- Slow acetylators are at higher risk of peripheral neuropathy.



Isoniazid

Adverse drug reactions: Less common

CNS toxicity

- **Seizure:** Low GABA → loss of CNS inhibition → hyperexcitability
 - symptom: generalized tonic-clonic seizures (overdose cases)
 - treatment: Intravenous pyridoxine (1 g of pyridoxine/g of INH ingested) → antidote for INH-induced seizures
- **Psychiatric symptoms:**
 - Neurotransmitter imbalance: decreased GABA & altered serotonin/dopamine metabolism.
 - symptoms: irritability, confusion, depression, restlessness (rare in therapeutic doses but possible in overdose)



Isoniazid

Adverse drug reactions: Less common

- Gastrointestinal upset:
Nausea, vomiting, epigastric pain, diarrhea
- Hypersensitivity reactions:
Rash, fever, or drug-induced lupus-like syndrome. (rare but can require discontinuation).
- Hematologic effects:
Agranulocytosis, thrombocytopenia, or anemia (rare)



Isoniazid

Drug interactions: Potent CYP2C19, CYP3A4 inhibitor, CYP2E1 Inducer, weak CYP2D6 inhibitor

SOME ISONIAZID-DRUG INTERACTIONS VIA INHIBITION AND INDUCTION OF CYPs

COADMINISTERED DRUG	CYP ISOFORM	ADVERSE EFFECTS
Acetaminophen	CYP2E1 induction	Hepatotoxicity
Carbamazepine	CYP3A inhibition	Neurological toxicity
Diazepam	CYP3A and CYP2C19 inhibition	Sedation and respiratory depression
Ethosuximide	CYP3A inhibition	Psychotic behaviors
Isoflurane and enflurane	CYP2E1 induction	Decreased effectiveness
Phenytoin and fosphenytoin	CYP2C19 inhibition	Neurological toxicity
Theophylline	CYP3A inhibition	Seizures, palpitation, nausea
Vincristine	CYP3A inhibition	Limb weakness and tingling
Warfarin	CYP2C9 inhibition	Possibility of increased bleeding (higher risk with isoniazid doses >300 mg/day)

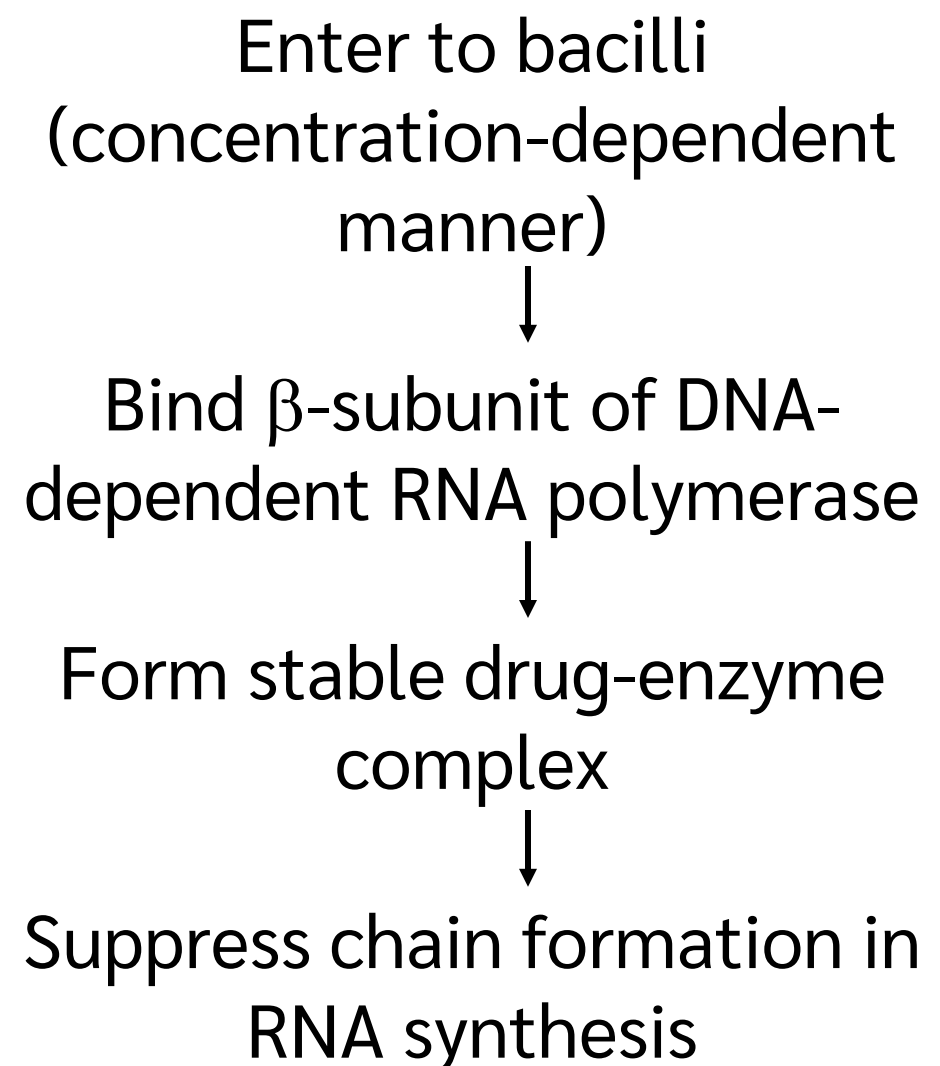


Rifamycins: Rifampin (Rifampicin), Rifapentine and Rifabutin

- RNA synthesis inhibitors
- Rifapentine & rifabutin & are rifampin analogues that have some activity against rifampin-resistant *Mycobacterium tuberculosis*.

Mechanism of action:

- inhibits DNA-dependent RNA polymerase.





Rifamycins: Rifampin (Rifampicin), Rifapentine, and Rifabutin

Resistance:

- Alteration of *rpoB* gene (encoding RNA polymerase)
 - 86% of cases due to mutation at codons 526 and 531 of the *rpoB* gene
 - commonly seen in multidrug-resistant TB (MDR-TB).
- Always used in combination therapy to prevent resistance.



Rifampin

- Has wide range of antibacterial activity.
 - inhibit growth of gram-positive bacteria
 - *Mycobacteria*, *Clostridium difficile*, *Staphylococcus aureus*
 - inhibit growth of gram-negative bacteria
 - *Neisseria meningitidis*, *N gonorrhoeae*, *Hemophilus influenza*
- Bactericidal inhibit growth of
 - *M. tuberculosis* (MIC: 0.06-0.25 mg/L)
 - *M. leprae*, *M. kansasii* (MIC: 0.25-1 mg/L)



Rifampin

Pharmacokinetics:

- **Absorption:** orally well absorbed (Food: ↓absorption 30%)
- **Distribution:** highly lipophilic, widely distributed throughout the body, crosses blood-brain barrier
- **Metabolism:** hepatic via deacetylation
- **Elimination:**
 - feces (60% to 65%), urine (~30%) as unchanged drug.
 - free drug & metabolites are orange-colored.



Rifampin

Adverse drug reactions:

- Generally well-tolerate
 - rash, fever, nausea, vomiting, diarrhea
- Hepatotoxicity
 - ↑ liver enzymes
 - ↑ risk when co-administration of isoniazid/pyrazinamide.
- Hypersensitivity reaction
 - fever, rash, urticaria, angioedema, hypotension, acute bronchospasm, conjunctivitis, or flu-like syndrome



Rifampin

Adverse drug reactions:

- Flu-like syndrome:
 - High incidence in dose of >600 mg once or twice weekly in adults.
- Discoloration:
 - Teeth (may be permanent), urine, feces, saliva, sweat, and tears may be discolored (yellow, orange, red, or brown).
 - “Red-man syndrome”: overdose
- Drug-induced thrombocytopenia (rare):
 - Bruising or bleeding



Rifampin

Drug interactions:

- Inducers
 - CYP3A4 (strong)
 - CYP2C19 (strong)
 - CYP2C9 (moderate)
 - CYP2B6 (moderate)
 - P-glycoprotein/ABCB1



Rifampin

Drug interactions:

Type of Drug	Comments
β -Adrenergic blocking agents	Monitor patient for clinical response; increased propranolol hydrochloride or metoprolol dose may be needed
Chloramphenicol†	Monitor serum chloramphenicol concentrations; may need to increase dosage
Clarithromycin	Monitor signs and symptoms of infection; more study needed
Dapsone	Monitor clinical response; dosage increase may be necessary; additional study needed when used for <i>Pneumocystis carinii</i> prophylaxis; monitor for hematologic toxic effects
Diazepam†	Monitor clinical response; may need to increase diazepam dosage
Digoxin (oral)	Monitor arrhythmia control and signs and symptoms of heart failure; monitor digoxin serum concentrations
Diltiazem†	Use alternative agent if possible because large oral doses of diltiazem may be ineffective; monitor clinical response‡
Disopyramide	Monitor arrhythmia control; increase dosage if needed
Doxycycline	Monitor clinical response; increase dosage if needed
Fluconazole	Monitor clinical response; may need to increase fluconazole dosage; less reduction in serum concentrations vs other azoles
Haloperidol†	Monitor clinical response; increase dosage if needed
Losartan potassium	Monitor patient for clinical response; may need to increase dosage
Nifedipine	Alternative class of agents should be considered; monitor clinical response; dosage increase may be needed†
Nortriptyline hydrochloride	Monitor clinical response and serum nortriptyline concentrations
Pefloxacin	Moderate rifampin induction effect; pending further research, no dosage adjustment recommended
Sulfonylureas	Monitor blood glucose levels; base any dosage adjustments on blood glucose control
Tacrolimus	Monitor serum tacrolimus concentrations and clinical response; increased dose may be needed or use another agent if possible
Tocainide	Monitor arrhythmia control; increase dosage if needed

*Data adapted from Baciewicz and coworkers,^{1,2} Borchering et al,³ and Strayhorn et al.⁴ Additional study is needed to clearly establish clinical significance. Carefully adjust doses when rifampin use is discontinued. The enzyme induction effect is gradually reduced during a 1- to 2-week period or longer.

†Probably of clinical significance.

‡See also data on verapamil in Table 1.



Rifampin

Drug interactions:

Type of Drug	Comments
Controlled Drug Interaction Studies	
Selective serotonin receptor (5-HT ₃) antagonist ^{44,45}	Monitor clinical response; increase dose if needed; use another agent if needed
Buspirone hydrochloride ^{11,12}	Monitor clinical response; increased dose will likely be needed or use another agent if possible
3-Hydroxy-3-methylglutaryl coenzyme A reductase inhibitors ^{17,18}	Monitor lipid panel; increased dose will likely be needed for simvastatin; further research needed for other agents in this class
Metronidazole ⁵⁵	Monitor for decreased clinical response; increase dose if needed or use another agent if possible
Opiates (morphine or codeine) ⁴⁶⁻⁴⁸	Monitor pain control and clinical response; increased dose may be needed in extensive metabolizers; use another agent if possible; may be associated with ethnic variability
Propafenone hydrochloride ^{20,21}	Monitor clinical response; increased dose may be needed or use another agent if possible
Tamoxifen citrate or toremifene citrate ⁴⁹	Monitor clinical response; increased dose likely needed
Zolpidem tartrate ¹⁴	Monitor clinical response; increased dose may be needed or use another agent if possible
Potential Interactions Based on Case Reports†	
Clozapine ¹³	Monitor clinical response; increase dose if needed or use another agent if possible
Levothyroxine sodium ⁵¹	Monitor thyrotropin level; increased dose likely needed
Sertraline hydrochloride ⁹	Monitor clinical response; increase dose if needed

*Carefully adjust dosage when rifampin use is initiated and discontinued. The enzyme induction effect is gradually reduced during a 1- to 2-week period or longer when rifampin therapy is discontinued. Based on the small number of reports, further studies are needed for most of these agents.

†Controlled study is needed to establish the importance and extent of the interaction.



Other Rifamycins

- **Rifapentine:**
 - Has a longer half-life, allowing once-weekly dosing with isoniazid for latent TB treatment.
 - Can be used once daily in combination with INH, pyrazinamide, and moxifloxacin for drug-susceptible pulmonary TB.
- **Rifabutin:**
 - Equally effective as rifampin for treating mycobacterial infections.
 - Preferred in AIDS patients due to fewer drug interactions, especially with cytochrome P450 substrates like protease inhibitors or efavirenz.



Pharmacokinetics

Table I. Comparisons of the absorption of rifampicin (rifampin), rifabutin and rifapentine

Drug (dose)	Bioavailability (%)	t_{max} (h)	C_{max} (mg/L)	Effect of food	Effect of antacids
Rifampicin 600mg	68	1.5-2.0	8-20	Decrease AUC 6%, decrease C_{max} 36%	None
Rifabutin 300mg	20	2.5-4.0	0.2-0.6	AUC and C_{max} unchanged, t_{max} increased	None (didanosine)
Rifapentine 600mg	Unknown	5-6	8-30	Increase AUC 40-50%	Unknown

AUC = area under the concentration-time curve; **C_{max}** = peak serum drug concentration; **t_{max}** = time to C_{max} .

Table II. Comparison of the elimination of rifampicin (rifampin), rifabutin and rifapentine and their interactions with cytochrome P450 (CYP) 3A

Interaction	Rifampicin	Rifabutin	Rifapentine	References
Major metabolic pathway	Deacetylation, hydrolysis to formyl derivatives	CYP3A-mediated hydroxylation, deacetylation	Deacetylation, hydrolysis to formyl derivatives	17,74,75
Serum half-life (h)	2-5	32-67	14-18	17-20,73
Effect on CYP3A	Pronounced	Weak	Moderate	76
Auto-induction	Yes	Yes	No	17,25,31
Example of CYP3A induction: effect on indinavir AUC	92% decrease	34% decrease	70% decrease	77,78
Change in AUC when given with a CYP3A inhibitor ^a	No effect	293% increase	No effect	77-79

a From interaction studies with ritonavir (rifampicin and rifabutin) and indinavir (rifapentine).

AUC = area under the concentration-time curve.

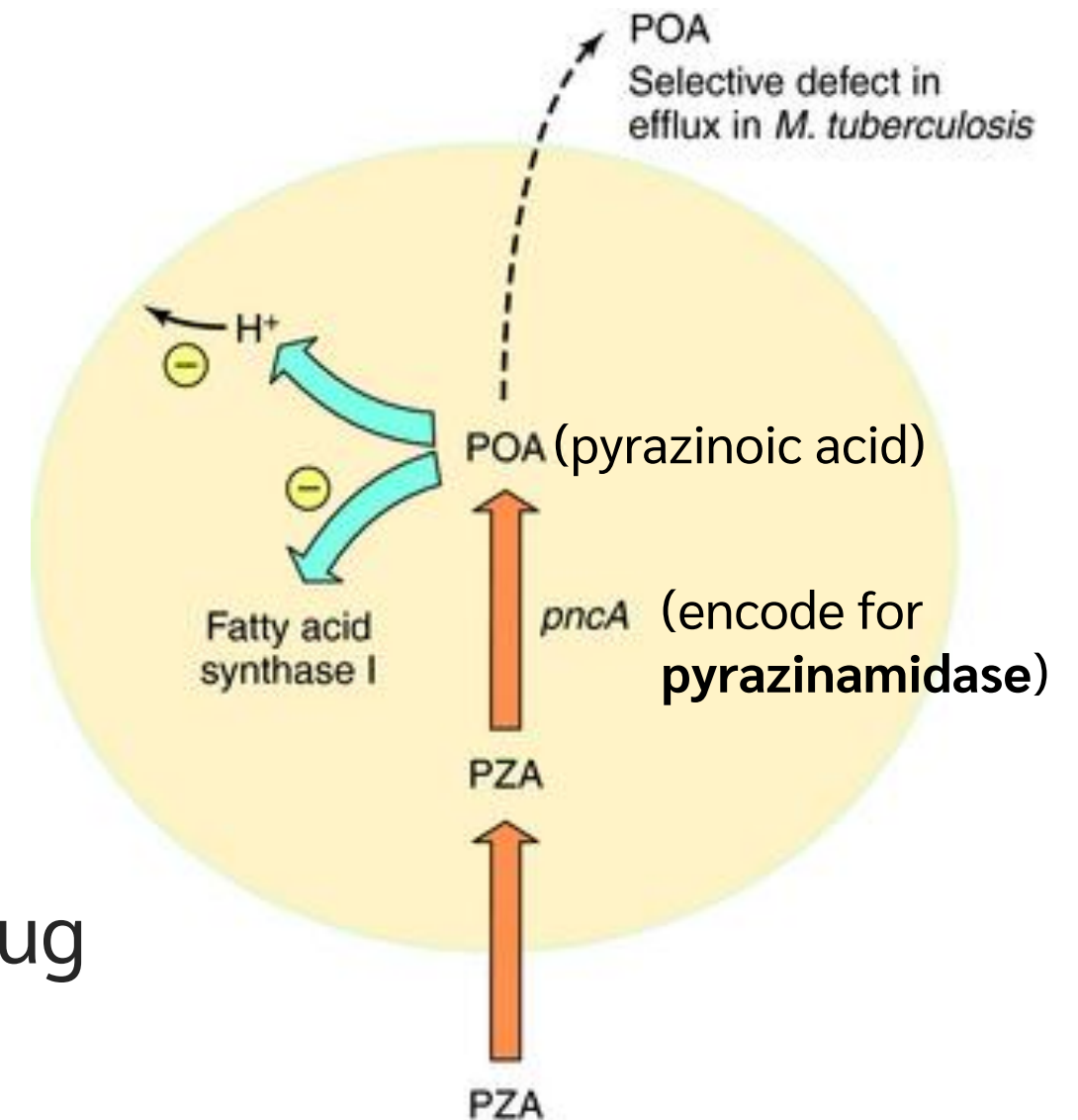
Pyrazinamide

Mechanism of action:

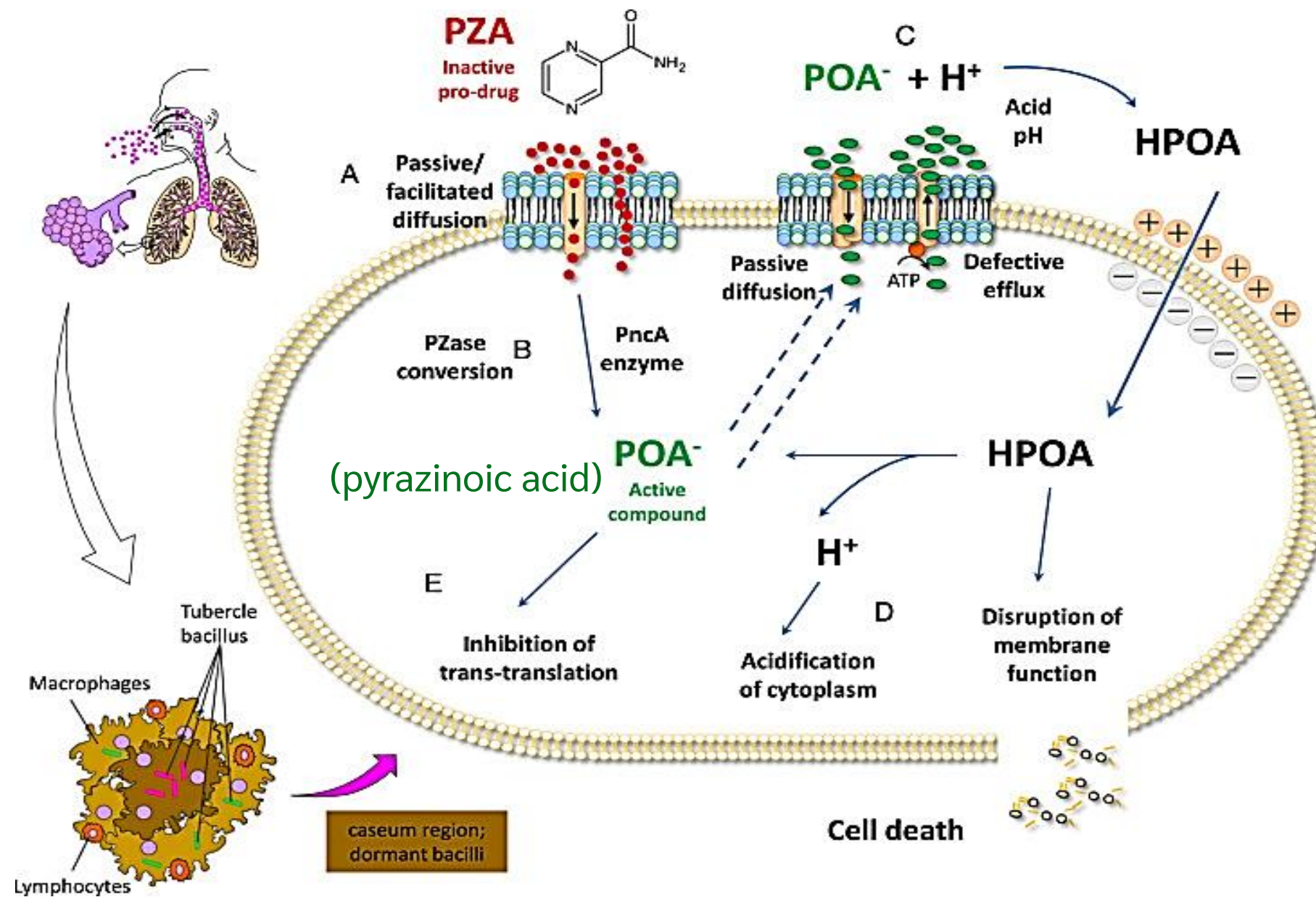
- inhibit fatty acid synthase type I
 - affects lipid metabolism by disrupting energy production
- exhibit antibacterial activity only at acidic pH (inactive at neutral pH).

Resistance:

- *pncA* gene mutation (reducing prodrug activation)
- Increase expression of drug efflux systems.



Pyrazinamide



- Damaging membrane potential.
- Interfering with energy production (ATP synthesis).



Pyrazinamide

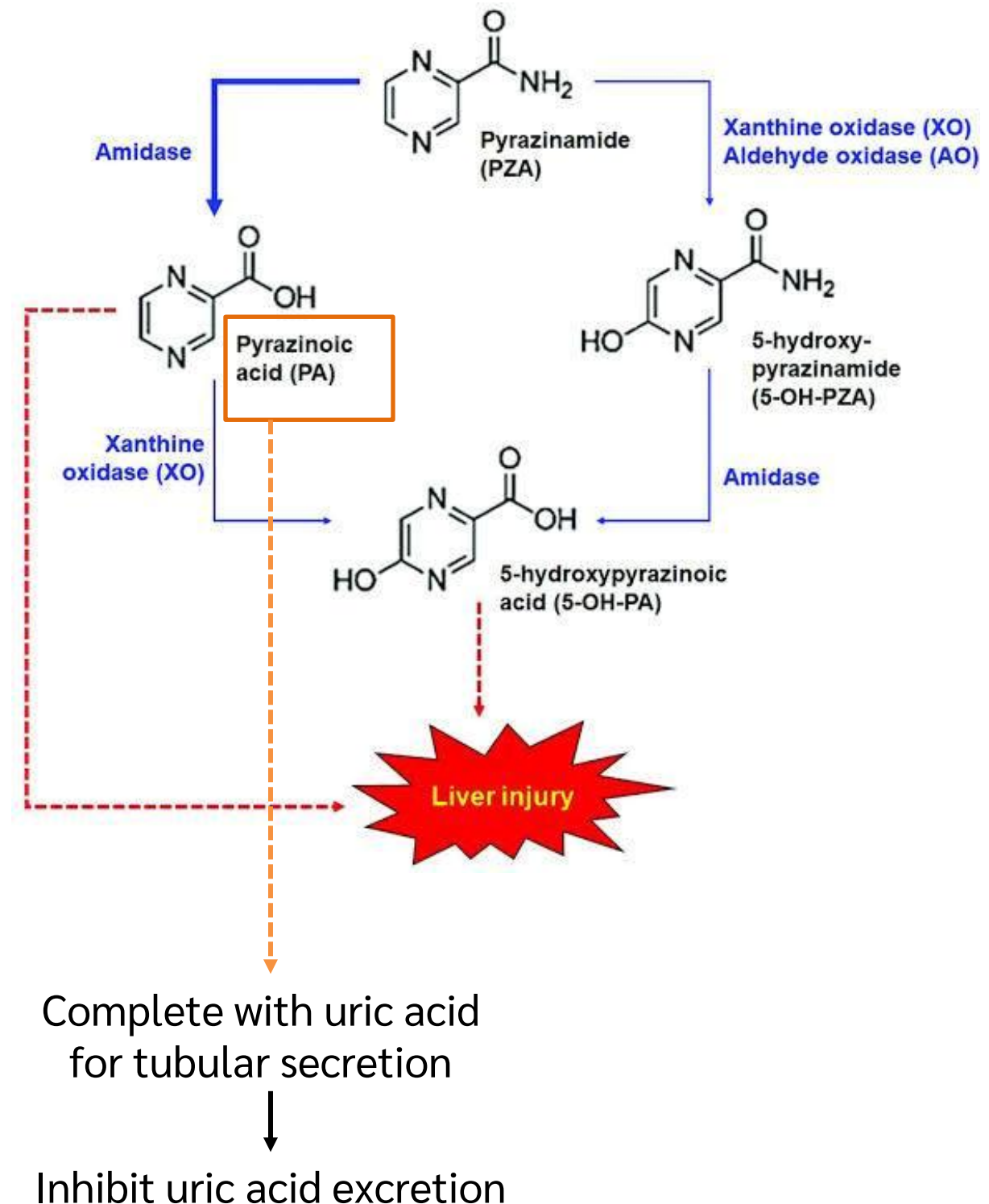
Pharmacokinetics:

- **Absorption:**
 - oral bioavailability: > 90%
- **Distribution:** widely into body tissues & fluids including liver, lung, and CSF.
- **Metabolism:** mainly hepatic via hydrolysis
- **Elimination:** Urine
 - Half-life elimination: 9 -10 hrs.
 - Pyrazinamide & metabolites are primarily excreted in urine.

Pyrazinamide

ADRs:

- Hepatotoxicity
- Hyperuricemia
- Arthralgia
- Others: anorexia, nausea, vomiting, dysuria, fever





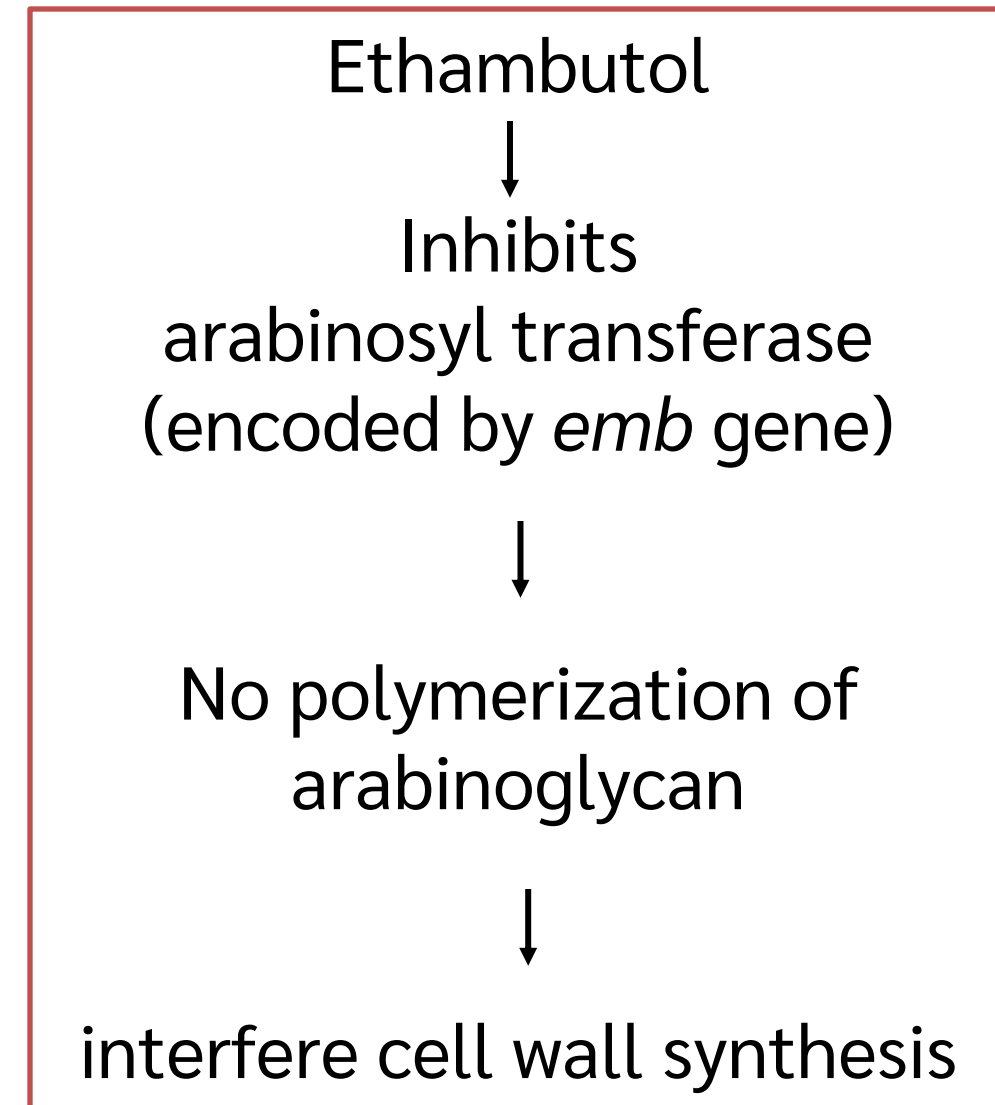
Ethambutol

Mechanism of action:

- Inhibits arabinosyl transferase resulting in impaired mycobacterial cell wall synthesis.

Resistance:

- *emb* gene mutation
- increase efflux pump activity





Ethambutol

Pharmacokinetics:

- **Absorption:** ~ 80%.
- **Distribution:** widely distributed throughout the body
- **Metabolism:** hepatic (20%) to inactive metabolite
- **Elimination:**
 - Urine (~50% as unchanged drug, ~ 8-15% as metabolites);
 - Feces (~20% as unchanged drug)



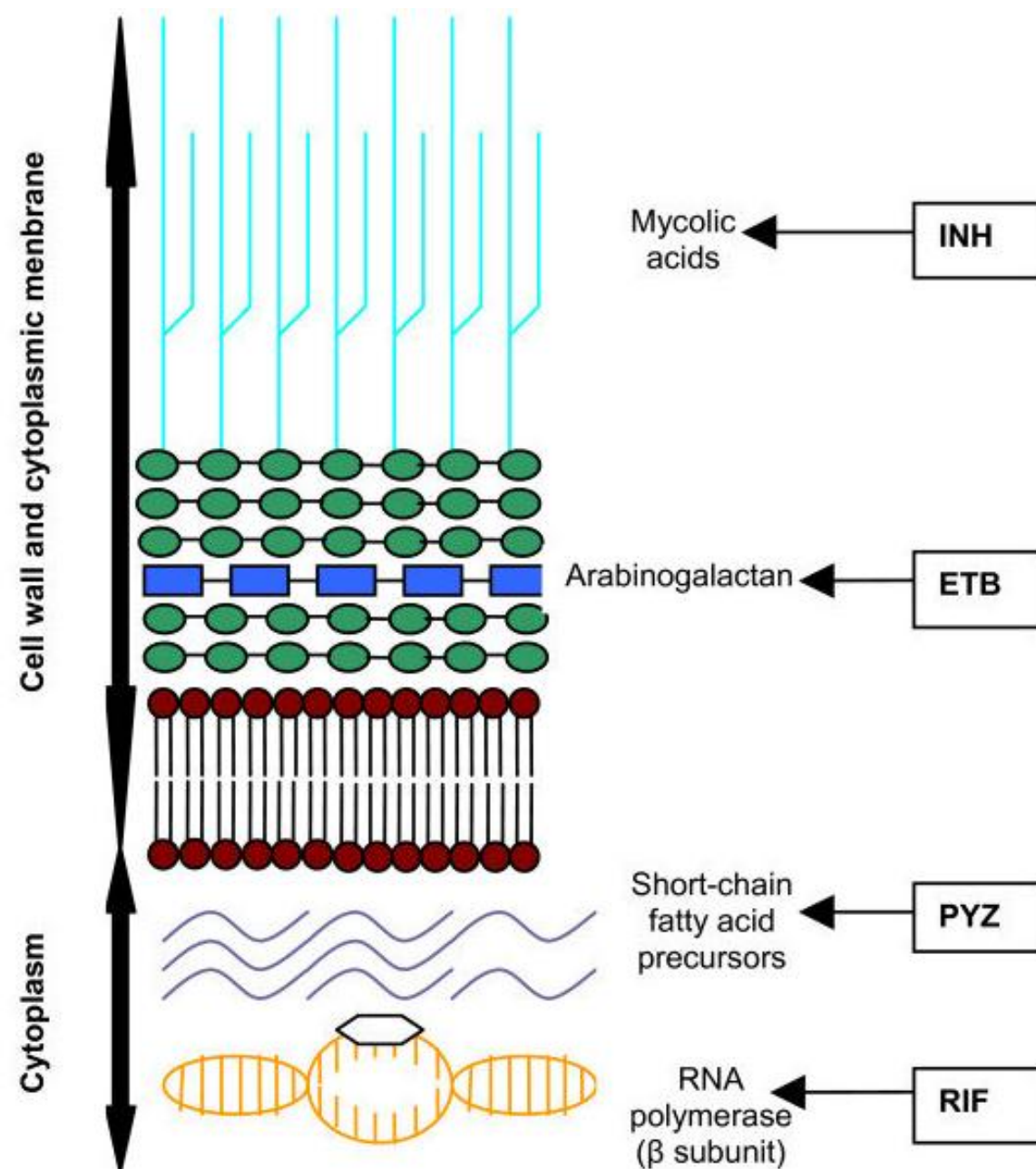
Ethambutol

Adverse drug reactions:

- Visual disturbances: “Optic neuritis ”
 - ↓ visual acuity, red-green color blindness, and possible retinal damage. (from prolonged use at high doses).
 - depend on duration & dose
 - recommend to test of visual acuity prior to start dosing.
 - Recovery usually occurs when ethambutol is withdrawn.
- Hepatotoxicity
- Others:
 - headache, confusion, joint pain, hyperuricemia, peripheral neuritis



Summary for 1st line Anti-TB therapy



Drugs	Mechanism of action	Mechanism of drug resistant	CYP-mediated activities	ADRs
Isoniazid (I, INH)	Inhibits cell wall mycolic acid Biosynthesis	Decreased drug Activation “resistant of <i>katG</i> , <i>inhA</i> gene	Potent CYP450 inhibitors :2C19,3A4 / weak inhibitor of CYP2D6	Peripheral, neuropathy, hepatitis (Treatment with vitamin B6 dose 50-100 mg)
Ethambutol (E)	Inhibits cell-wall Arabinogalactan synthesis	Target modification “resistant of <i>embB</i> gene	not significant	Optic neuritis, decreased visual acuity, hyperuricemia
Pyrazinamide (Z)	Inhibits protein translation and membrane energetics	Decreased drug activation “resistant of <i>pncA</i> gene”	not significant	Gastrointestinal (GI) upset, rash, joint pain, hyperuricemia
Rifamycins (R) rifampin, rifapentine, rifabutin	Binds to RNA polymerase inhibiting RNA synthesis	Target modification “resistant of <i>rpoB</i> gene”	Potent CYP450 inducers:1A2, 2C8,2C9,2C19, 3A4	GI upset, rash, hepatitis, flu-like symptoms, red-man syndrome (sign of overdose)

Recommended dose for first-line anti-TB drugs in adults

น้ำหนัก ก่อนเริ่มการรักษา (กก.)	ขนาดของยา			
	H (มก.)** (4-6 มก./กก./วัน)	R (มก.) (8-12 มก./กก./วัน)	Z (มก.) (20-30 มก./กก./วัน)	E (มก.) (15-20 มก./กก./วัน)
35*-49	300	450	1,000	800
50-69	300	600	1,500	1000
> 70*	300	600	2,000	1,200

หมายเหตุ * ในกรณีน้ำหนัก < 35 หรือ > 70 กิโลกรัม ให้คำนวณขนาดยาตามน้ำหนักตัว

** H สามารถปรับตามน้ำหนักตัว และชนิด Acetylator gene ของผู้ป่วย (NAT2 genotype)

ตารางที่ 5.2 อาการไม่พึงประสงค์จากยารักษาวัณโรคแนวที่หนึ่ง และการรักษา



ADRs of First-line anti-TB drugs

อาการไม่พึงประสงค์รุนแรง	ยาที่เป็นสาเหตุได้	แนวทางการดำเนินการรักษา
อาการรุนแรงหรือมีแนวโน้มที่จะรุนแรง		หยุดยาที่อาจเป็นสาเหตุได้ และส่งปรึกษาผู้เชี่ยวชาญ
ผื่นคันและผื่นที่ไม่มีอาการคัน	H, R, Z	หยุดยารักษาวัณโรค
ภาวะดีซ่าน (ที่ไม่พบสาเหตุอื่น) ตับอักเสบ	H, z, R	หยุดยารักษาวัณโรค
อาการสับสน (ถ้าเกิดร่วมกับอาการตับ อักเสบให้สงสัยยาเป็นสาเหตุ)	ยารักษาวัณโรค เกือบทุกตัว	หยุดยารักษาวัณโรค
การมองเห็นบกพร่อง (ที่ไม่พบสาเหตุอื่น)	E	หยุดยา ethambutol
ภาวะช็อค ผื่นผิวหนังชนิด purpura ไตวายเฉียบพลัน	R	หยุดยา rifampicin
อาการไม่รุนแรง		สามารถให้ยาวัณโรคต่อได้ โดยตรวจสอบขนาดยา
เบื่ออาหาร คลื่นไส้ ปวดท้อง	Z, R, H	บริหารยาโดยการกินยาร่วมกับอาหารปริมาณ น้อย หรือกินยาก่อนนอน แนะนำให้ผู้ป่วยกลืนยาช้าๆ และจิบน้ำตามเล็กน้อย ถ้ายังมีอาการหรืออาการแย่ลง เช่น อาเจียนพุ่ง หรือมีเลือดออกร่วมด้วย ให้พิจารณาเป็นผลข้าง เคียงแบบรุนแรง และส่งต่อผู้เชี่ยวชาญเพื่อรักษา อาการเร่งด่วน
ปวดข้อ	Z	ให้การรักษาด้วยยากลุ่ม NSAID หรือ ยา paracetamol
อาการแสบ ชา เจ็บปลายมือปลายเท้า	H	ให้การรักษาด้วยยา pyridoxine ขนาด 50-75 มก. ต่อวัน
ง่วงนอน	H	ให้ยาก่อนนอน
ปัสสาวะสีส้มแดง	R	
อาการคล้ายไข้หวัดใหญ่	R	พิจารณาหยุดยาถ้าอาการรุนแรง



Second-line anti-TB drugs

Second-line therapy is required when:

- **Non-compliance**
- **Drug interaction with other drugs:** Isoniazid, rifampin
- **Treatment intolerance:**
 - Patients unable to tolerate first-line drugs due to severe adverse effects (e.g., hepatotoxicity or hypersensitivity).

Side effects	Drugs
Rash	H,R,Z,E
Hepatitis	H,R,Z
Optic neuritis	E, H
Thrombocytopenia	R
Peripheral neuropathy	H, R
Flu-like symptoms	R

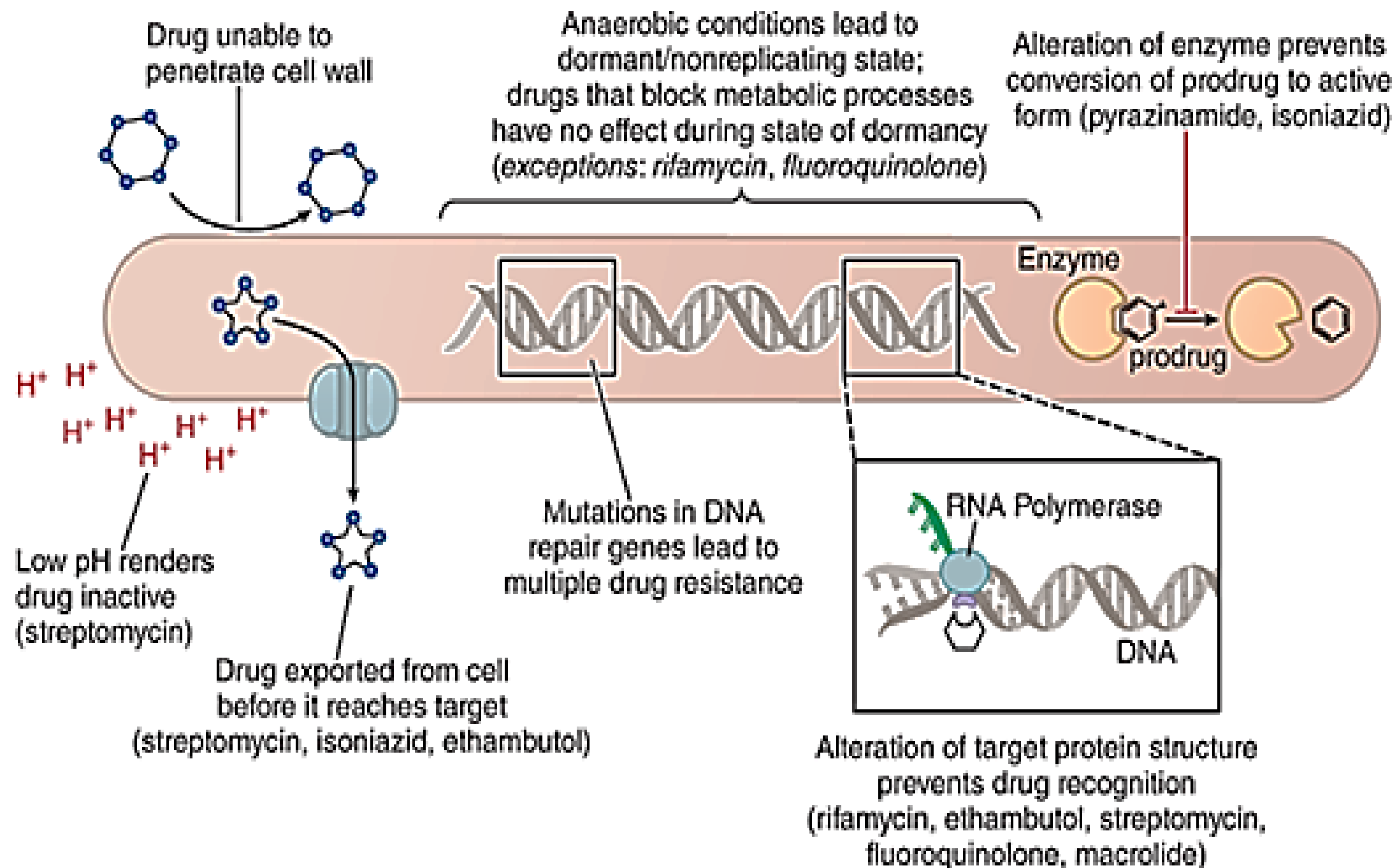


Second-line anti-TB drugs

Second-line therapy is required when:

- **Drug-resistant TB:**
 - Rifampicin-resistant TB (RR-TB)
 - Multidrug-Resistant TB (MDR-TB): Resistance to isoniazid & rifampin.
 - Extensively Drug-Resistant TB (XDR-TB): MDR-TB with additional resistance to fluoroquinolones & at least one second-line injectable drug.
- **Relapse or treatment failure:**
 - TB re-emerges or persists despite proper adherence to first-line regimens.
- **Special populations:**
 - In certain high-risk cases such as HIV co-infection.

Resistance mechanisms of mycobacteria





Second-line anti-TB drugs

- **Aminoglycosides:**
 - streptomycin, amikacin, kanamycin, capreomycin
- **Fluoroquinolones:**
 - moxifloxacin, gatifloxacin, levofloxacin, ofloxacin
- **Oral bacteriostatic anti-TB drugs:**
 - ethionamide, cycloserine, para-aminosalicylic acid (PAS)
- **New anti-TB drugs:**
 - bedaquiline, delamanid, protomanid



Aminoglycosides

- Streptomycin, kanamycin, amikacin, capreomycin
- **MOA:** bind to the 30S ribosomal subunit → inhibit bacterial protein synthesis
- Administer via IM (streptomycin, capreomycin) or IV (amikacin, kanamycin) routes
- **Resistance:**
 - *rrs* gene mutation (16S rRNA)
 - ↑ expression of efflux pump



Aminoglycosides

- **Pharmacokinetics:**
 - Poor oral absorption
 - Minimally or no hepatic metabolism
 - Eliminate as unchanged via glomerular filtration
- **ADRs:**
 - Nephrotoxicity
 - Ototoxicity



Fluoroquinolones

- Moxifloxacin, gatifloxacin, levofloxacin, ofloxacin
- **MOA:** disrupt the DNA-DNA gyrase complex → inducing double-strand DNA cleavage → block DNA synthesis
- **Resistance:**
 - Target modification in the *gyrA* or *gyrB* genes.
- **Pharmacokinetics:**
 - Oral bioavailability: 80-90%
 - Chelation interaction: ↓ Absorption if taken with multivalent cations (antacids with Al, Mg, or supplements with Calcium, Iron, Zinc)
 - Metabolism & elimination
 - Ciprofloxacin, levofloxacin, ofloxacin: mainly renal excretion
 - Moxifloxacin: mainly hepatic metabolism → no renal dose adjustment

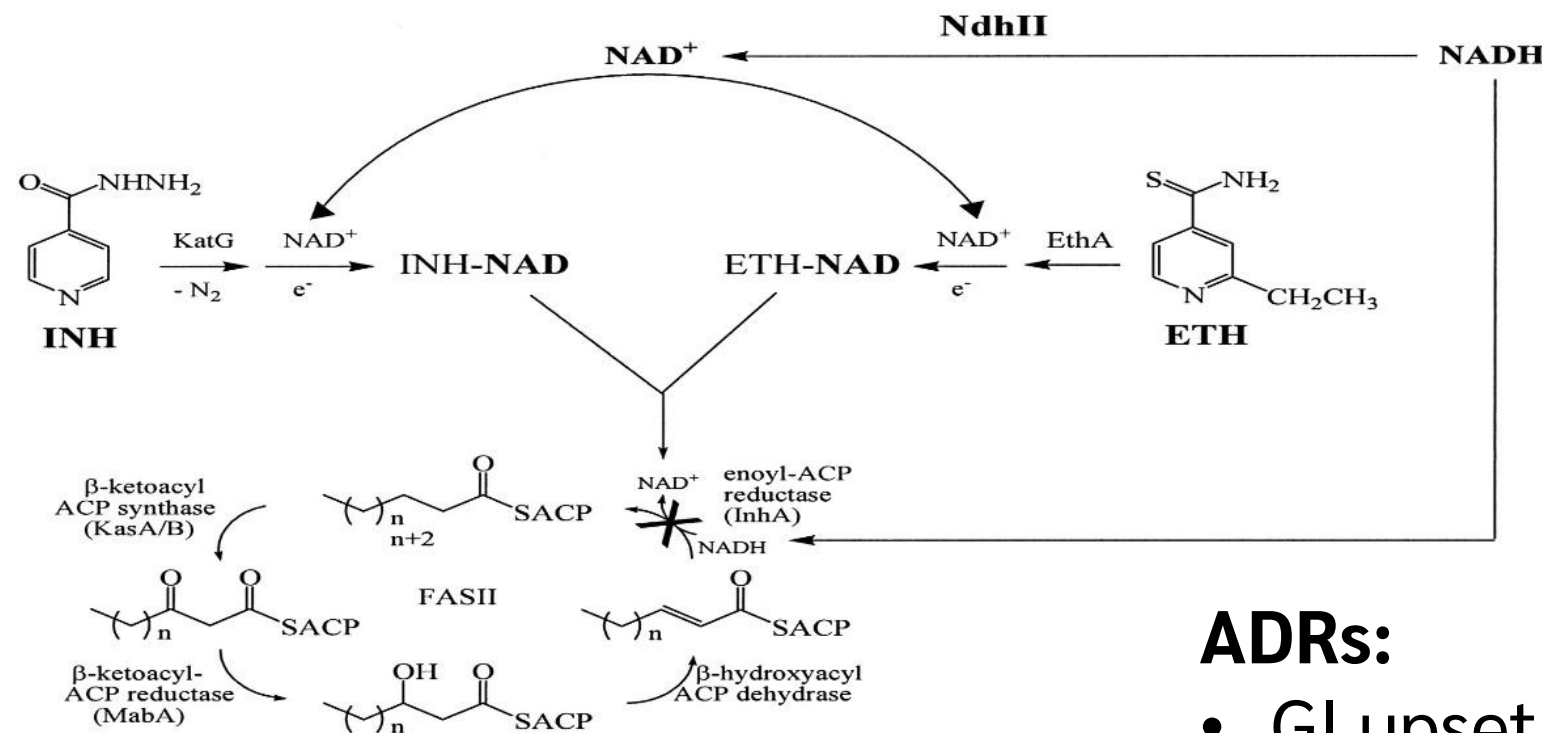


Fluoroquinolones

- **ADRs:**
 - Gastrointestinal (GI) upset
 - Tendonitis
 - QT prolongation
 - Peripheral neuropathy
 - CNS effects (dizziness, confusion, seizures)
 - Dysglycemia
 - Photosensitivity

Ethionamide

MOA: Inhibit mycolic acid biosynthesis and impaired cell wall synthesis, bacteriostatic effect



- Used in MDR-TB as part of second-line regimens.

ADRs:

- GI upset (anorexia, n/v, gastric irritation)
- Postural hypotension
- Mental distress
- Hepatitis (5%)

Genes with alleles that transfer resistance or susceptibility to INH or ETH upon transformation

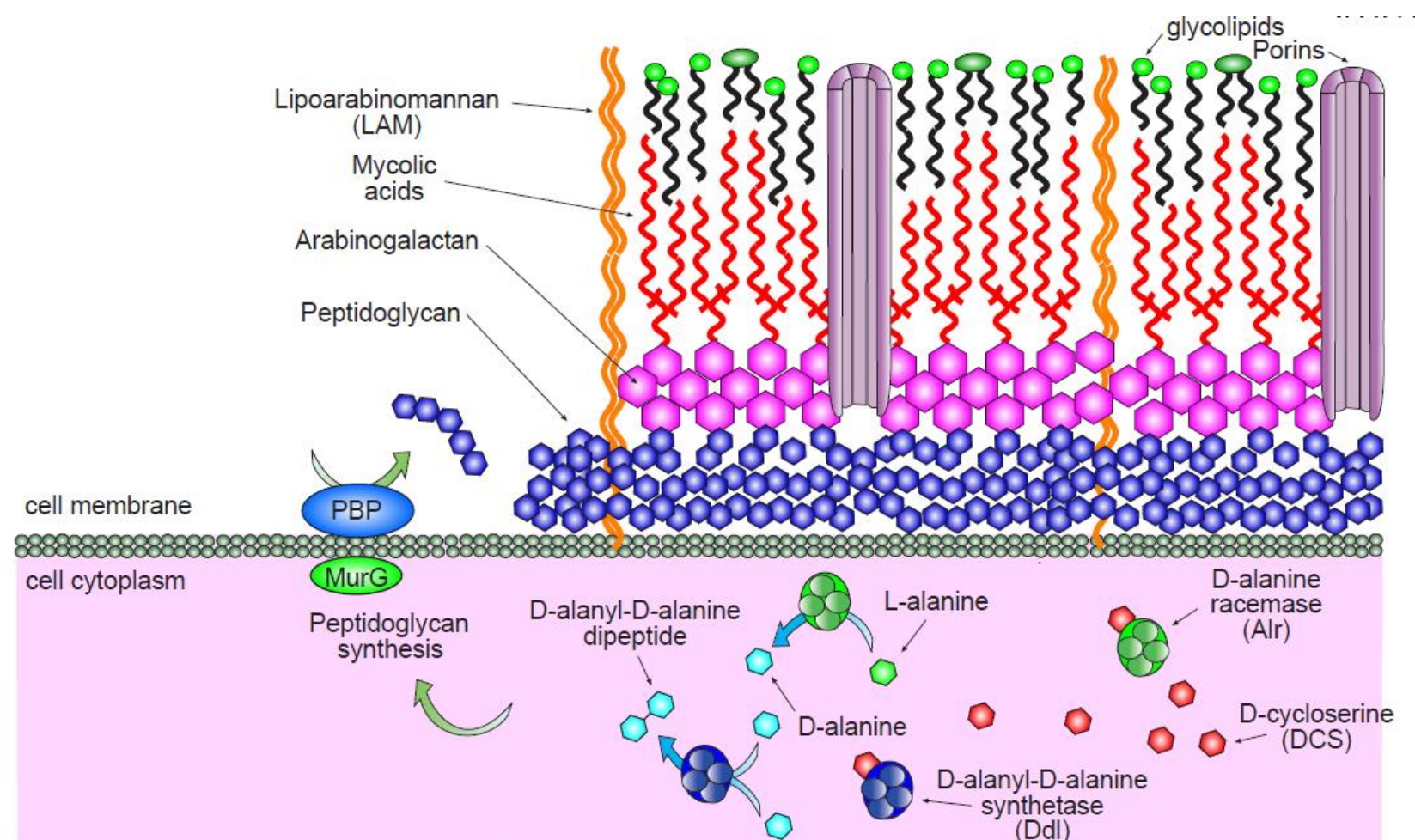
	INH resistance	ETH resistance
<i>katG</i>	+	-
<i>ethA</i>	-	+
<i>inhA</i>	+	+
<i>ndh</i>	+	+

Palmer et al., *Pharmaceuticals* 2012, 1147-1159

Cycloserine

MOA: inhibit 2 enzymes essential for peptidoglycan synthesis.

- D-alanine racemase(Alr): converts L-alanine to D-alanine.
- D-alanyl-D-alanine synthetase (Ddl): combines 2 D-alanine molecules for incorporation into the cell wall.





Cycloserine

ADRs:

- Neurotoxicity (common)
 - anxiety, depression, confusion, irritability.
 - severe cases: psychosis, seizures, suicidal ideation.
- Others:
 - Headache, dizziness, insomnia
 - Peripheral neuropathy (rare)
 - Hematologic (rare): anemia or leukopenia in isolated cases.

Para-aminosalicylic acid (PAS)

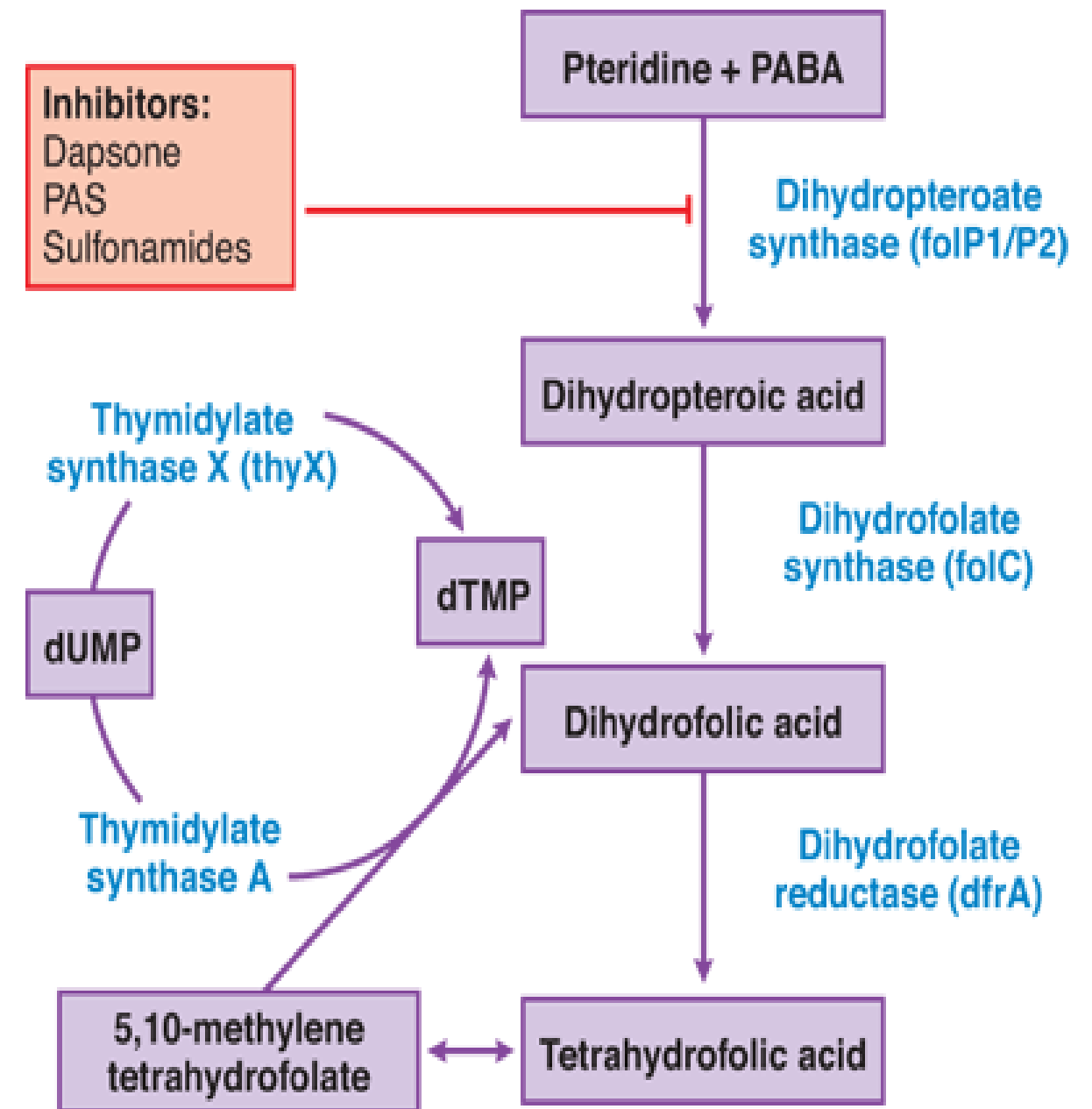
MOA: Inhibit folate synthesis & inhibit synthesis of cell wall component, bacteriostatic effect

- Liver: PAS → → → N-acetyl PAS (hepatotoxin)
- **Resistance:** mutation in thymidylate synthase gene (*thyA*)

Indications: Used in MDR-TB treatment as part of second-line regimens.

ADRs:

- GI upset (common), hypersensitivity, skin eruption, fever, eosinophilia, hematological abnormality





Bedaquiline

- **MOA:** inhibits mycobacterial ATP synthase, an enzyme essential for energy production in *Mycobacterium tuberculosis*.
 - *unique MOA among TB drugs, making it effective against both active & dormant bacteria.*
- **Use:**
 - Part of BPaLM regimen (Bedaquiline, Pretomanid, Linezolid, Moxifloxacin) for treating XDR-TB & treatment-intolerant MDR-TB.
- **ADRs:**
 - QT prolongation (risk of cardiac arrhythmias)
 - Nausea, headache, and hepatotoxicity (rare)

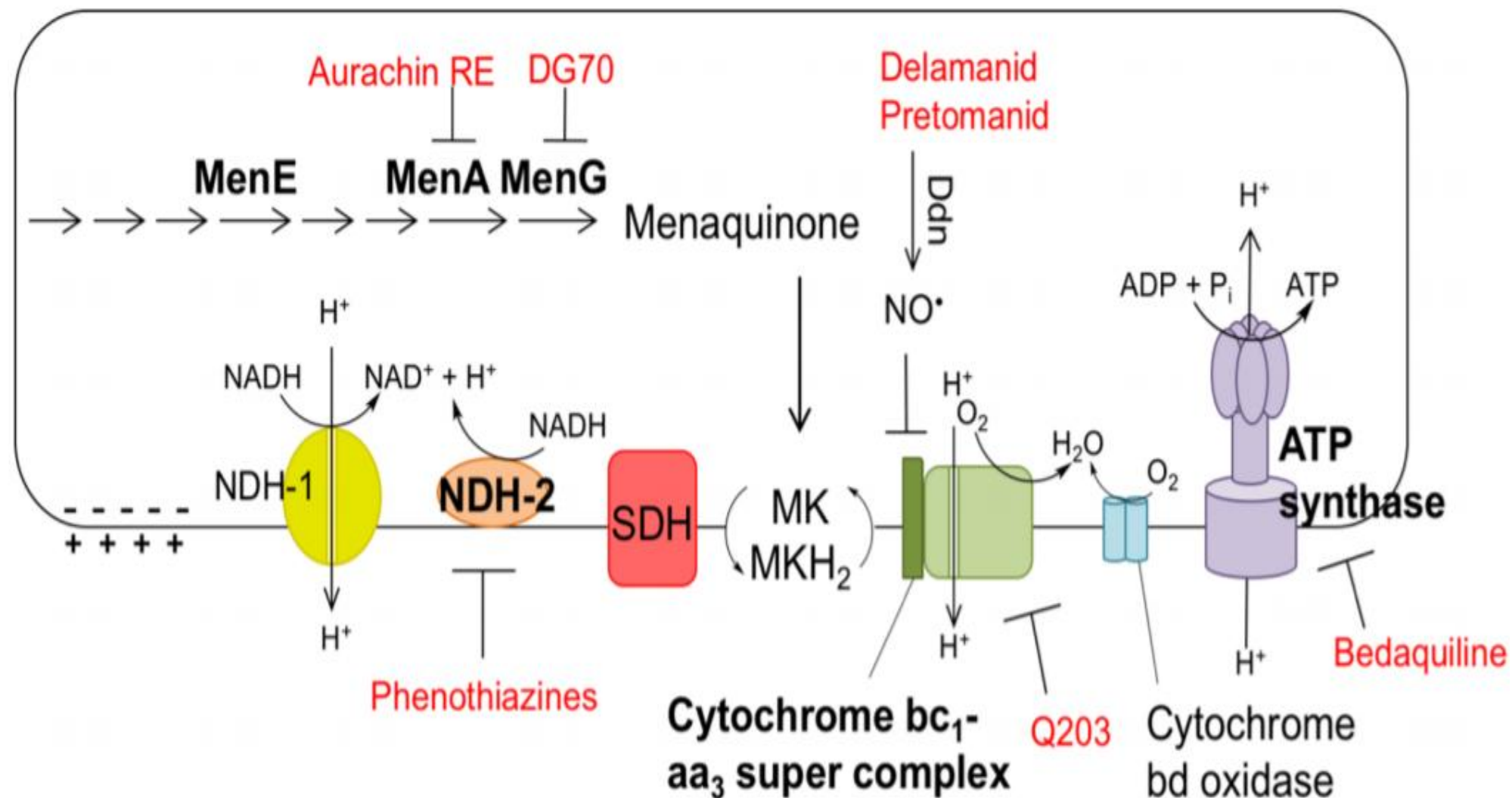


Bedaquiline

Pharmacokinetics:

- Absorption: Bioavailability is enhanced in presence of food.
- Plasma protein binding: >99%
- Long half-life: 24-30 hrs.
- Metabolism: primarily by CYP3A4 to N-monodesmethyl metabolite (4-6 times less potent than the parent drug)
 - Co-administration with CYP3A4-inducers/inhibitors should be avoided!
- Elimination: mainly in feces

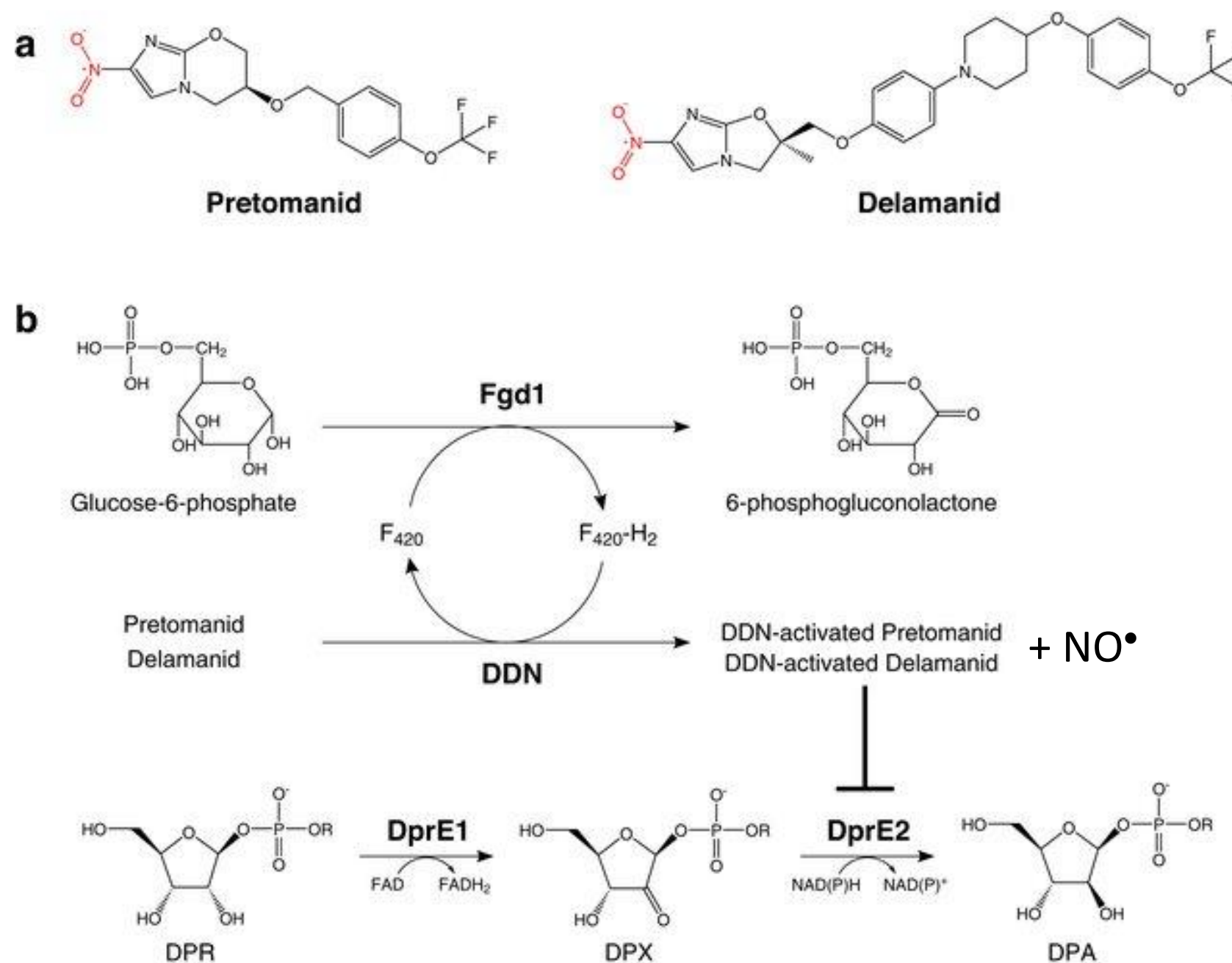
Protomanid/Delamanid



ACS Infect. Dis. 2018;4: 696–714

- Disrupt bacterial energy production by targeting ATP synthesis & respiration.
- Induce oxidative & nitrosative stress, damaging cellular components like DNA, proteins, and lipids.
- Enhance bactericidal activity against non-replicating (dormant) bacteria.

Protomanid/Delamanid



DDN: deazaflavin dependent nitroreductase



Protomanid

- **MOA:**
 - inhibiting mycolic acid synthesis
 - generating reactive nitrogen species that are toxic to the bacteria.
- **Use:**
 - Part of BPaLM regimen for treating XDR-TB and treatment-intolerant MDR-TB.
- **ADRs:**
 - Neuropathy, myelosuppression, gastrointestinal upset.



Protomanid

Pharmacokinetics:

- Absorption: oral bioavailability is increased when taken with a high-fat meal.
- Plasma protein binding: ~ 86%
- Half-life: ~ 16-20 hrs.
- Metabolism:
 - metabolized by CYP3A4,
 - avoid co-administration with strong CYP3A4 modulators (CYP3A4 inducers e.g., rifampicin or inhibitors e.g., ketoconazole)
- Elimination:
 - feces (53%), urine (36%) as inactive metabolites



Delamanid

- **MOAs:** inhibits mycolic acid synthesis, crucial components of the mycobacterial cell wall & generates reactive nitrogen species, enhancing its bactericidal activity.
- **Use:**
 - in combination regimens for MDR-TB, especially where bedaquiline is unavailable or contraindicated.
- **ADRs:**
 - QT prolongation, gastrointestinal disturbances, and hepatotoxicity.



Delamanid

Pharmacokinetics:

- Absorption: oral bioavailability is low.
- Plasma protein binding: >99%
- Half-life: ~ 30-38 hrs.
- Metabolism:
 - primarily by plasma albumin through non-cytochrome P450-dependent pathways.
 - minor metabolized by CYP3A4
- Elimination: mainly in feces



Linezolid

- **MOA:** binds to the 23S portion of the 50S subunit of the bacterial ribosome → inhibits formation of initiation complex → inhibit bacterial protein synthesis.
- **Pharmacokinetics:**
 - Oral bioavailability: ~ 100%
 - Distribution: 30-50 L, good tissue penetration into skin, bone, fat, lung, CSF
 - Plasma protein binding: >31%
 - Half-life: ~ 5-7 hrs.
 - Metabolism: hepatic (non-CYP enzymes)
 - Elimination: primarily excreted in urine



Linezolid

- **ADRs:**
 - Myelosuppression (esp. thrombocytopenia with prolonged use)
 - Peripheral and optic neuropathy (with long-term use)
 - Lactic acidosis (rare but serious)
 - Serotonin syndrome (when combined with SSRIs or MAO inhibitors)
 - GI upset: nausea, vomiting, diarrhea

ตารางที่ 2.1 กลุ่มยาที่ใช้ในการรักษาวัณโรคดื้อยา

กลุ่มยา	ยา	
Group A: Include all three medicines (unless they cannot be used)	Levofloxacin <u>OR</u>	Lfx
	Moxifloxacin	Mfx
	Bedaquiline	Bdq
	Linezolid	Lzd
Group B: Include both medicines (unless they cannot be used)	Clofazimine	Cfz
	Cycloserine <u>OR</u>	Cs
	Terizidone	Trd
Group C: Add to complete the regimen and when medicines from Groups A and B cannot be used	Ethambutol	E
	Delamanid	Dlm
	Pyrazinamide	Z
	Imipenem-cilastatin <u>OR</u>	Ipm-Cln
	Meropenem	Mpm
	Amikacin (<u>OR</u> Streptomycin)	Am (S)
	Ethionamide <u>OR</u>	Eto
	Prothionamide	Pto
	p-aminosalicylic acid	PAS

ที่มา : World Health Organization. WHO consolidated guidelines on tuberculosis Module 4: Treatment drug-resistant tuberculosis treatment. Geneva: World Health Organization; 2020.

ตารางที่ 6.7 อันตรกิริยาระหว่างยาวัณโรคและยาอื่นๆ

ยา	คู่ยาที่เกิดอันตรกิริยาร่วมกัน	คำแนะนำ
ยารักษาวัณโรคคือยา กลุ่ม A-C		
กลุ่ม A		
Levofloxacin Moxifloxacin Gatifloxacin	- การใช้ยาที่มีส่วนประกอบของประจุบวก เช่น aluminium, magnesium จะทำให้ลดการดูดซึมของ fluoroquinolones	- รับประทานยาห่างกันอย่างน้อย 2 ชั่วโมง
	- อาจเพิ่มฤทธิ์ของยา warfarin	- ควรติดตามค่า INR และ prothrombin time อย่างใกล้ชิด
	- เพิ่มระดับ theophylline ในกระแสเลือด	- ติดตามระดับยา theophylline ในเลือด
Bedaquiline	CYP3A4 inhibitors (กลุ่มยาฆ่าเชื้อรา azole, macrolides บางตัว, protease inhibitors และอื่นๆ) อาจทำให้เพิ่มระดับยา bedaquiline ในกระแสเลือด	ควรหลีกเลี่ยงการใช้ยาร่วมกัน
	ยา rifampicin (เป็น CYP3A4 inducer) จะทำให้ลดระดับยา bedaquiline ในเลือดลงครึ่งหนึ่ง	ควรหลีกเลี่ยงการใช้ยาร่วมกัน
	หลีกเลี่ยงการใช้ที่ทำให้เกิด QT prolongation เช่น clofazimine, fluoroquinolones, delamanid, ยาฆ่าเชื้อรากลุ่ม azole	ตรวจติดตามวัด ECG อย่างสม่ำเสมอ

ยา	คู่ยาที่เกิดอันตรกิริยาร่วมกัน	คำแนะนำ
Linezolid	- การให้ยาร่วมกับ buspirone, mereridone, fluoxetine และยาในกลุ่ม serotonin 5-HT1 antagonists ทำให้เกิด serotonin syndrome	- หลีกเลี่ยงการให้ร่วมกัน
กลุ่ม B		
Clofazimine	Clofazimine เป็น weak inhibitor ของ CYP3A4 clofazimine อาจชะลอการดูดซึมของ rifampicin และ ทำให้ time to Cmax นานขึ้น มีรายงานการเกิด Drug Interactions กับ dapsone, oestrogen และ vitamin A	- ควรติดตามผลการรักษา อย่างใกล้ชิด ข้อแนะนำ ควรรับประทาน ยานี้กับอาหารที่มีไขมันสูง เพิ่ม Cmax และ AUC
Cycloserine	- Ethionamide และ isoniazid จะเพิ่มอาการ ช้างเคียงทาง CNS มากขึ้น - การดื่มแอลกอฮอล์ร่วมกับยา Cycloserine ทำให้เพิ่มอาการชักได้	- ให้วิตามินบีหก - งดดื่มแอลกอฮอล์ ในระหว่างการได้รับยานี้
กลุ่ม C		
Delamanid	หลีกเลี่ยงการใช้ที่ทำให้เกิด QT prolongation	ตรวจติดตามวัด ECG อย่างสม่ำเสมอ
	ยา metabolized ผ่าน cytochrome P450 enzyme like CP3A4 เนื่องจากเป็นยาใหม่จึงไม่มี clinical significant ในการใช้ยานี้ร่วมกับกลุ่ม protease inhibitors	หลีกเลี่ยงการให้ร่วมกัน
Amikacin Capreomycin	- การใช้ยากลุ่มนี้ร่วมกับยา amphotericin B, cephalosporin, cyclosporin, cisplatin, furosemide และ vancomycin อาจทำให้พิษต่อ หูและไตมากขึ้น	- ไม่แนะนำให้ใช้ร่วมกัน
Capreomycin	- เพิ่มฤทธิ์ของยา neuromuscular blocking agents โดยเฉพาะ non-depolarizing เช่น vecuronium	- ปรับขนาดยา non-depolarizing muscle relaxant อย่างช้าๆ และ ติดตามการทำงานของ neuromuscular function อย่างใกล้ชิด

ยา	คู่ยาที่เกิดอันตรกิริยาร่วมกัน	คำแนะนำ
Ethionamide Prothionamide	- การให้ยาร่วมกับ cycloserine อาจจะทำให้เกิด neurotoxicity เพิ่มขึ้นและทำให้เกิดการชักเพิ่มขึ้นด้วย - การดื่มแอลกอฮอล์ร่วมกับยามีผลต่อภาวะทางจิต - การให้ยาร่วมกับ PAS อาจจะทำให้ความเป็นพิษต่อตับ - ภาวะ hypothyroidism อาจเกิดขึ้นได้ในกรณีที่ให้ร่วมกันมากกว่าการใช้ยาเดี่ยว	- ให้วิตามินบีหก ขนาดสูงสุด (200 mg/วัน) - ถ้าอาการไม่ดีขึ้น ควรหยุดยา และเริ่มกลับเข้าไปใหม่ในขนาดยาต่ำก่อน - งดดื่มแอลกอฮอล์ในระหว่างการได้รับยานี้ - ติดตามค่า liver enzymes
PAS	- ลดระดับยา digoxin - การให้ยาร่วมกับ ethionamide อาจจะทำให้ความเป็นพิษต่อตับ - ภาวะ hypothyroidism อาจเกิดขึ้นได้ในกรณีที่ให้ร่วมกัน - ลดระดับยา isoniazid เนื่องจากลดการขจัดลง (acetylation)	- ติดตามค่าระดับยา digoxin - ติดตามค่า liver enzymes - ติดตามค่าไทรอยด์ฮอร์โมน (T3,T4, TSH) อาจให้ยา thyroxine (levothyroxine) - อาจจะต้องเพิ่มขนาดยา

หมายเหตุ :

- ยากลุ่ม Fluoroquinolone ไม่ควรทานร่วมกับนมและผลิตภัณฑ์จากนม ยาลดกรด และอาหารหรือยาที่มีส่วนประกอบของเหล็ก แมกนีเซียม แคลเซียม และสังกะสี เนื่องจากลดการดูดซึมยา ดังนั้นควรรับประทานยาห่างกันอย่างน้อย 2 ชั่วโมง
- Linezolid ออกฤทธิ์เป็นตัวยับยั้งเอนไซม์ monoamine oxidase (MAO) อย่างอ่อนๆ ควรระวังการใช้ร่วมกับยาในกลุ่ม selective serotonin reuptake inhibitors หรือยาอื่นๆ ที่มีผลต่อ serotonin เนื่องจากเสี่ยงต่อการเกิด serotonin syndrome



Antimycobacterial drugs

Drugs used in Tuberculosis

- Isoniazid (H,INH)
- Rifampin (R)
- Pyrazinamide (Z)
- Ethambutol (E)
- Aminoglycosides
- Fluoroquinolones
- Ethionamide
- Para-aminosalicylic acid (PAS)
- Cycloserine
- Clofazimine

Drugs used in Leprosy

- Dapsone
- Clofazimine
- Rifampin

Drugs used in non-tuberculous mycobacteria

- Macrolides
- Fluoroquinolones
- Aminoglycosides



Leprosy (Hansen's Disease)

- Chronic bacterial infection by *Mycobacterium leprae*.
- *Predominantly affects the skin, peripheral nerves.*

Paucibacillary (PB): 1–5 skin lesions, without demonstrated presence of bacilli in a skin smear.

Multibacillary (MB) : > 5 skin lesions or with nerve involvement (pure neuritis, or any number of skin lesions & neuritis); or with the demonstrated presence of bacilli in a skin smear.

Treatment: Multidrug Therapy (MDT) Regimens

- Rifampin
- Clofazimine
- Dapsone

(PB – 6 months, MB – 12 months)



Clofazimine

- Both antibacterial activity & anti-inflammatory effects.
- First licensed for use in leprosy.
- **Possible MOA:**
 - DNA interaction: binds to bacterial DNA, disrupting replication and transcription.
 - Reactive oxygen species (ROS): promotes ROS generation within the bacterial cell → cell death
- **Indications:**
 - Leprosy (Hansen's Disease)
 - MDR-TB or XDR-TB treatment



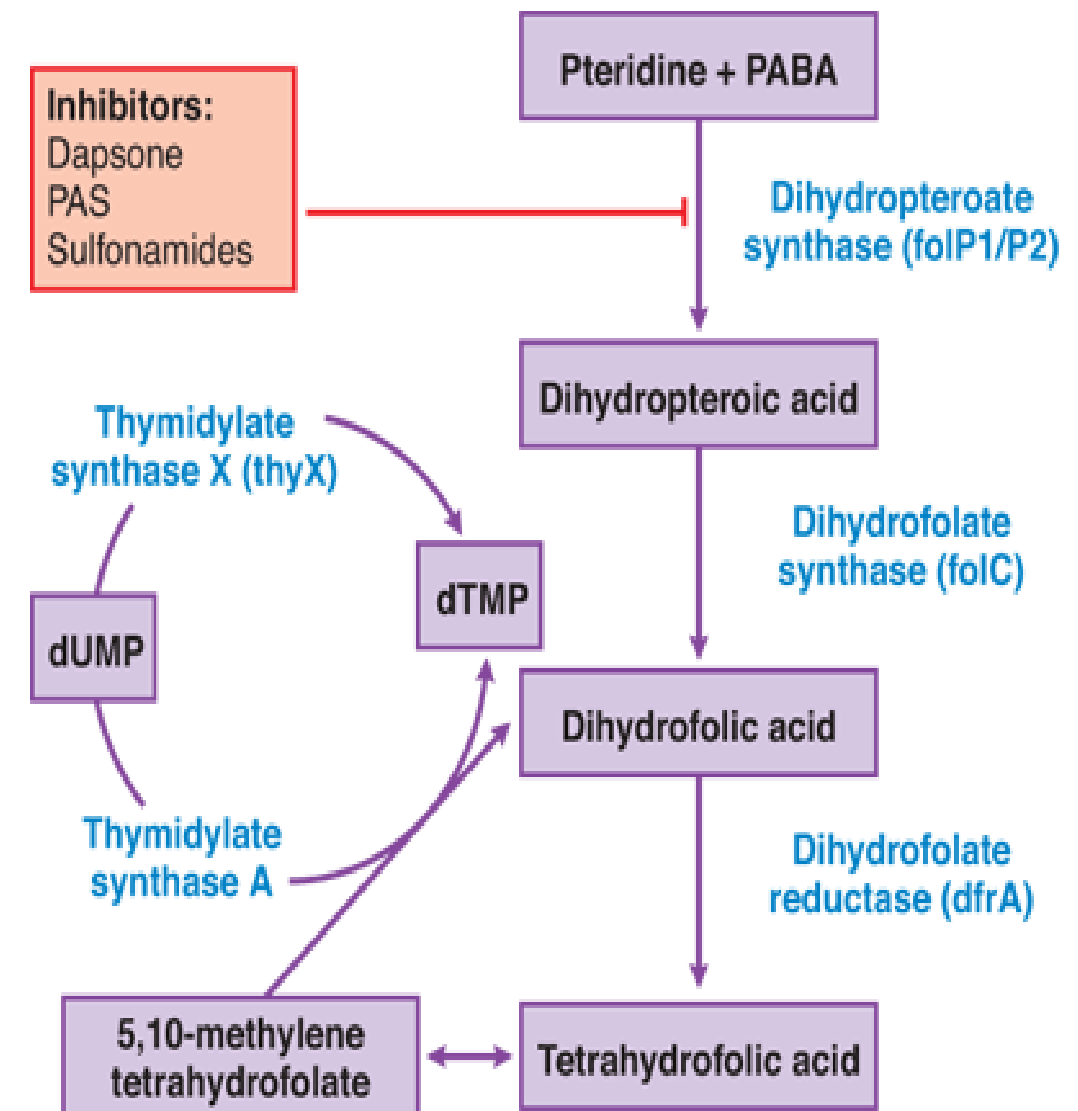
Clofazimine

ADRs:

- Skin discoloration: most common
 - Causes red to brown-black pigmentation of the skin, especially with long-term use.
 - May take months to years to resolve after discontinuation.
- Gastrointestinal symptoms:
 - Nausea, vomiting, diarrhea, and abdominal pain.
- Photosensitivity:
 - Increased risk of sunburn

Dapsone

- Broad spectrum agents
 - antibacterial
 - antiprotozoa (*Plasmodium falciparum*, *Toxoplasma gondii*)
 - antifungal (*Pneumocystic jiroveci*)
- **MOA:** inhibits dihydropteroate synthase, an enzyme involved in bacterial folic acid synthesis. -> disrupts DNA synthesis -> bacteriostatic against *M. leprae*





Dapsone

Pharmacokinetics:

- Absorption: well absorbed orally (70–80% bioavailability)
- Distribution: Widely distributed, crosses the placenta and enters breast milk.
- Half-life: ~ 20-30 hrs.
- Metabolism:
 - metabolized in liver through N-acetylation (NAT2)
- Elimination:
 - primarily via urine (85%)



Dapsone

ADRs:

- Hematological:
 - Hemolysis, especially in patients with G6PD deficiency.
 - Methemoglobinemia.
- Dermatological:
 - Rash, exfoliative dermatitis, and rarely dapsone hypersensitivity syndrome.
- Neurological:
 - Peripheral neuropathy at high doses
- Hepatotoxicity:
 - Rare but requires monitoring.



Mycobacterium Avium Complex (MAC)

- Made up of at least 2 species: *M. intracellulare* (pulmonary disease) , *M. avium*
- Found in the environment, including soil, water, dust.
- Often infects immunocompetent patients (opportunistic infections)
- Triple therapy:
 - rifamycin 600 mg
 - ethambutol 25mg/kg
 - macrolide (clarithromycin 1000 mg or azithromycin 500mg)



Drugs used in the treatment of mycobacteria other than TB, leprosy, MAC

Mycobacteria species	1 st line therapy	Alternative agents
<i>M. kansasii</i>	INH+rifampin+ethambutol	Trimethoprim-sulfonamide,ethiomide, cycloserine, clarithromycin
<i>M. fortuitum</i> complex	Amikacin+doxycycline	Cefoxitin, rifampin, sulfonamide, clarithromycin, sulfamethoxazole-trimethoprim
<i>M. marinum</i>	Rifampin+ethambutol	sulfamethoxazole-trimethoprim,clarithromycin, minocycline,doxycycline
<i>M. malmoense</i>	Rifampin+ethambutol± clarithormycin	fluoroquinolone
<i>M. haemophilum</i>	Clarithromycin+rifampin+ quinolone	



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Thank you